3:1

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1. 1
   1.1. 1
   1.2. 1
   1.3. 2

2. - , 4
   2.1. 3:1 - 4
   2.2. 4
   2.3. 5
   2.4. 6
   2.5. 8

3. 9
   3.1. 9
   3.2. 9
   3.3. 10
   3.4. 10

4. 10
   4.1. 11
      4.1.1. 11
      4.1.2. 14
   4.2. 16
      4.2.1. 16
      4.2.2. 17
   4.3. 18
      4.3.1. W^s W^u 20

5. 25

6. 25
   6.1. 1 25
   6.2. 26

1.

1.1.

3:1 . .
: (S), (J) (A), , $m_A \ll m_J \ll m_S$. S J , A , SJ.
[?]:

$$H = \frac{p_x^2 + p_y^2}{2} + p_x y - p_y x + U(x, y, t),$$

$U(x, y, t)$ - , $r(t)$ S J.
$e_J > 0$ (0.05) . , . (S J , $r(t) = \text{const}$, $e_J = 0$), , (Wisdom, 1987, "Chaotic behavior in the Solar System Meyer, Hall, 2017, "Introduction to Hamiltonian Dynamical Systems and the N-Body Problem").

1.2.

$$\nu = \frac{m_J}{m_S + m_J} \approx 10^{-3},$$

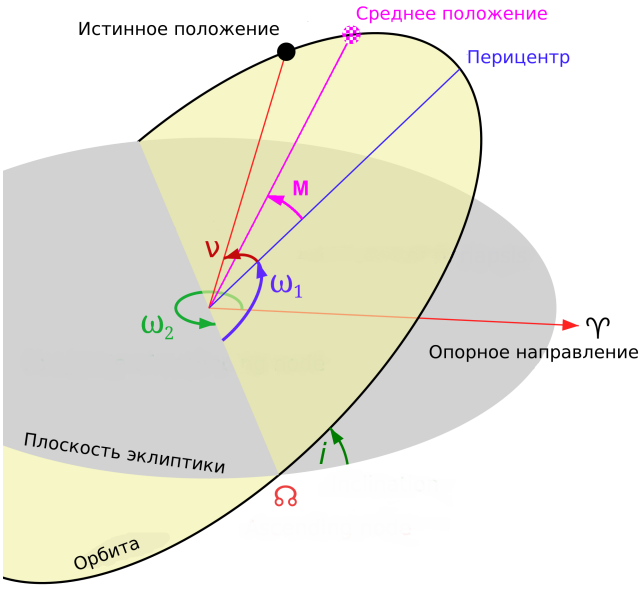
SJ.
(-) (MD, Plummer (1918) An Introductory Treatise on Dynamical Astronomy), [?]:

$$H = -\frac{(1-\nu)^2}{2L^2} - \nu R(L, \rho_1, \rho_2, l, \omega_1, \omega_2),$$

R :

$$R = \sum_{s-j-k+p-m-n=0, s\geq 0} K^{sjkpmn}(L, \rho_1, \rho_2) \cos(sl + pl_J + j\omega_1 + k\omega_2 + m\omega_{1J} + n\omega_{2J}).$$

L, ρ_1, ρ_2 : a, e i. – l, ω_1 ω_2. , , J.



. 1. $l = \omega_1 + \omega_2 + M$, M -

, , [?] ( ):

$$s\dot{l} + p\dot{l}_J + j\dot{\omega}_1 + k\dot{\omega}_2 + m\dot{\omega}_{1J} + n\dot{\omega}_{2J} \approx 0.$$

, l , :

$$s\dot{l} + p\dot{l}_J \approx 0 \qquad \Leftrightarrow \qquad \dot{l} \approx \frac{-p}{s}\dot{l}_J \equiv \frac{s+q}{s}\dot{l}_J.$$

q 3:1 $q = 2,\ s = 1$.
, ($i = 0,\ \rho_2 = 0,\ \omega_2 = 0$), e_J, [?]:

$$H = -\frac{(1-\nu)^2}{2L^2} - \nu R_{sec}(\rho_1, \omega_1) - \nu R_{res}(L, l, \rho_1, \omega_1),$$

$R_{sec}(\rho_1, \omega_1) = -2\rho_1 F - e_J G\sqrt{2\rho_1}\cos\omega_1$ - , ,
$R_{res}(L, l, \rho_1, \omega_1) = 2\rho_1 C\cos(l - \omega_1 - 3l_J) + e_J^2 E\cos(l - 3l_J)$ - , .
[?]:

$$a_J = 1, \dot{l}_J = 1.$$

:

$$\Lambda = L - L_{res}, \quad \lambda = l - 3l_J, \quad x = \sqrt{2\rho_1}\cos\omega_1, \quad y = \sqrt{2\rho_1}\sin\omega_1,$$

Λ, :

$$H = \alpha\frac{\Lambda^2}{2} + \nu\left(C(x^2 - y^2) + e_J Dx + e_J^2 E\right)\cos\lambda + \nu(2Cxy + e_J Dy)\sin\lambda + \nu e_J Gx + \nu F(x^2 + y^2), \quad (1)$$

$L_{res} = \left(\frac{(1-\nu)^2}{3}\right)^{\frac{1}{3}}$ - , :

$$\dot{\lambda} = \frac{\partial H}{\partial L}|_{L=L_{res}} = 0,$$

a α:

$$\alpha = -\frac{3(1-\nu)^2}{L_{res}^4} < 0.$$

, y λ , x L . , λ, , :

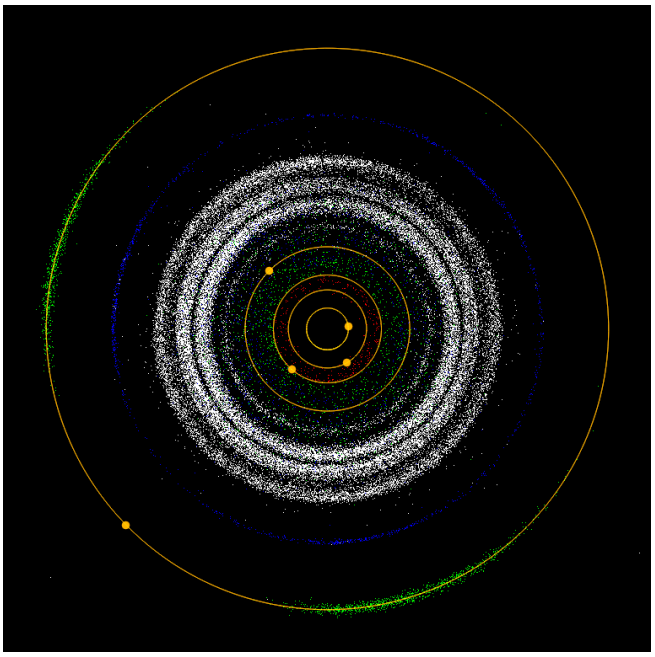
$$(x, y, \Lambda, \lambda) \in \mathbb{R}^3 \times S^1.$$

F, G, C, D, E, e_J , - , , [?].
, (1), 3:1.

1.3.

– , . 1866 . . , . : 2:1, 3:1, 5:2, 7:3.

. 2.



. 3.

. , . [?]:

$$\begin{cases} \frac{d\Lambda}{dt} = \nu\big(U(x,y)\sin\lambda - V(x,y)\cos\lambda\big), \\ \frac{d\lambda}{dt} = \alpha\Lambda, \\ \frac{dx}{dt} = -\nu\left(2Fy + \frac{\partial U(x,y)}{\partial y}\cos\lambda + \frac{\partial V(x,y)}{\partial y}\sin\lambda\right), \\ \frac{dy}{dt} = \nu\left(2Fx + e_J G + \frac{\partial U(x,y)}{\partial x}\cos\lambda + \frac{\partial V(x,y)}{\partial x}\sin\lambda\right), \end{cases} \tag{2}$$

$$U(x,y) = C(x^2 - y^2) + e_J Dx + e_J^2 E,$$
$$V(x,y) = 2Cxy + e_J Dy.$$

$\frac{dx}{dt}$ $\frac{dy}{dt}$ ν, x y ν^{-1}, λ ($\nu^{-\frac{1}{2}}$). , x y - : $x = \overline{x} + \xi$, $y = \overline{y} + \eta$, :

$$\begin{cases} \frac{d\overline{x}}{dt} = -\nu\left(2F\overline{y} + \frac{\partial U(\overline{x},\overline{y})}{\partial \overline{y}} < \cos\lambda > + \frac{\partial V(\overline{x},\overline{y})}{\partial \overline{y}} < \sin\lambda >\right), \\ \frac{d\overline{y}}{dt} = \nu\left(2F\overline{x} + e_J G + \frac{\partial U(\overline{x},\overline{y})}{\partial \overline{x}} < \cos\lambda > + \frac{\partial V(\overline{x},\overline{y})}{\partial \overline{x}} < \sin\lambda >\right), \end{cases} \tag{3}$$

T:

$$< \cos\lambda >= \frac{1}{T}\int_0^T \cos\lambda dt = \begin{cases} \frac{2E(k_L)}{K(k_L)} - 1, & -\sqrt{U^2+V^2} < H < \sqrt{U^2+V^2}, \\ \frac{2E(k_C)}{k_C^2 K(k_C)} + 1 - \frac{2}{k_C^2}, & H < -\sqrt{U^2+V^2}, \end{cases}$$

$$< \sin\lambda >= \frac{1}{T}\int_0^T \sin\lambda dt = 0,$$

$$k_L = \sqrt{\frac{\nu\sqrt{U^2+V^2} - H}{2\nu\sqrt{U^2+V^2}}}, \quad k_C = \sqrt{\frac{2\nu\sqrt{U^2+V^2}}{\nu\sqrt{U^2+V^2} - H}},$$

$K(k)$ - 1 , $E(k)$ - 2 .
1 , - .
, . , λ $\overline{x}$ $\overline{y}$. $\frac{d\lambda}{dt} = O(\nu)$, , $\frac{dx}{dt}$ $\frac{dx}{dt}$ ν. , $U(x,y)\sin\lambda - V(x,y)\cos\lambda$ () .

2. - ,

2.1. 3:1 -

, (1) - . $\Lambda = \sqrt{\nu}\Lambda_{\text{new}}$, $t_{\text{new}} = \sqrt{\nu}t$. (“new”) $\varepsilon = \sqrt{\nu}$, (t):

$$\begin{cases} \dot{\Lambda} = U\sin\lambda - V\cos\lambda, & \dot{\lambda} = \alpha\Lambda, \\ \dot{x} = -\varepsilon\big(2Fy + \dfrac{\partial U}{\partial y}\cos\lambda + \dfrac{\partial V}{\partial y}\sin\lambda\big), & \dot{y} = \varepsilon\big(2Fx + e_J G + \dfrac{\partial U}{\partial x}\cos\lambda + \dfrac{\partial V}{\partial x}\sin\lambda\big). \end{cases} \tag{4}$$

λ Λ , x y .

2.2.

(4) :

$$M_0 = \{(x, y, \lambda, \Lambda) : \quad \Lambda = 0, \quad u(x, y)\sin\lambda = v(x, y)\cos\lambda\}.$$

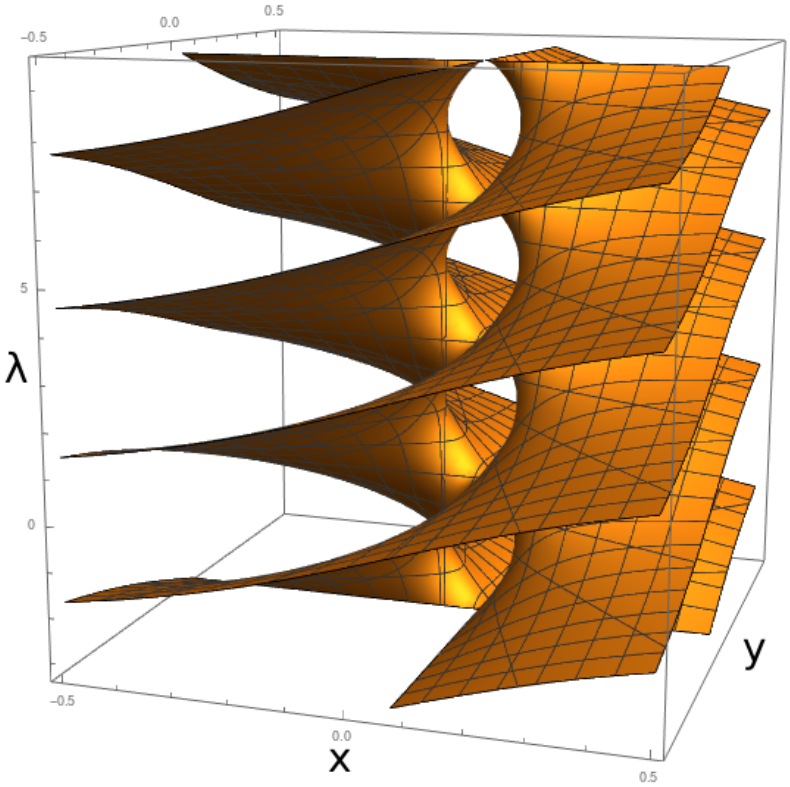
$U\sin\lambda = V\cos\lambda$ $\lambda = \arctan(V/U) + \pi k$, $k \in \mathbb{Z}$. $U = 0$ $\lambda_\pm : \mathbb{R}^2 \setminus \{U = V = 0\} \to S^1$, π:

$$\lambda_+(x, y) \equiv \begin{cases} \arctan\frac{V}{U}, & U > 0, V > 0, \\ \arctan\frac{V}{U} + \pi, & U < 0, \\ \arctan\frac{V}{U} + 2\pi, & U > 0, V < 0, \end{cases} \qquad \lambda_-(x, y) \equiv \begin{cases} \arctan\frac{V}{U} + \pi, & U > 0, V > 0, \\ \arctan\frac{V}{U} + 2\pi, & U < 0, \\ \arctan\frac{V}{U} + 3\pi, & U > 0, V < 0. \end{cases}$$

, $U = V = 0$, :

$$y = 0, \quad x = x_b^\pm \equiv \frac{-e_J D \pm e_J\sqrt{D^2 - 4EC}}{2C}, \quad \Lambda = 0, \quad \lambda \in S^1.$$

$D^2 - 4EC > 0$ $\mathbb{R}^3 \times S^1$.
, , , $\tan\lambda = y/(x - x_b^\pm)$.



. 4.

, $M_{0,\pm}$, $M^{\pm}_{0,b}$:

$$
\begin{aligned}
M_0 &= M_{0,+} \cup M_{0,-} \cup M^{+}_{0,b} \cup M^{-}_{0,b},\\
M_{0,\pm} &= \{\lambda = \lambda_{\pm}(x,y), \Lambda = 0, (x,y) \in \mathbb{R}^2 \setminus \{U = V = 0\}\},\\
M^{\pm}_{0,b} &= \{y = 0, x = x^{\pm}_b, \Lambda = 0, \lambda \in S^1\}.
\end{aligned}
$$

$M_{0,\pm}$, $M^{\pm}_{0,b}$ - .

2.3.

(4) ε , "" :

$$
\begin{cases}
\dot{\Lambda} = U(x,y)\sin\lambda - V(x,y)\cos\lambda, & \dot{\lambda} = \alpha\Lambda,\\
\dot{x} = 0, & \dot{y} = 0.
\end{cases}
\tag{5}
$$

(x,y) U, V . λ t $\dot{\Lambda}$, :

$$\ddot{\lambda} - \alpha U \sin\lambda + \alpha V \cos\lambda = 0,$$

($\alpha < 0$) :

$$\ddot{\lambda} - \alpha\beta \sin\big(\lambda - \lambda_+(x,y)\big) = 0, \quad \beta(x,y) \equiv \sqrt{U^2 + V^2} > 0.$$

, ( ) "" , $(\lambda, \Lambda) = (\lambda_{\pm}(x,y), 0)$. (5) :

$$\begin{pmatrix} U\sin\lambda - V\cos\lambda \\ \alpha\Lambda \end{pmatrix} = \begin{pmatrix} 0 & \alpha \\ \pm\beta(x,y) & 0 \end{pmatrix} \begin{pmatrix} \lambda - \lambda_{\pm} \\ \Lambda \end{pmatrix} + O((\lambda - \lambda_{\pm})^2, \Lambda^2).$$

:

$M_{0,+}$:

$$\zeta = \pm i\sqrt{|\alpha|\beta},$$

$M_{0,-}$:

$$\zeta = \pm\sqrt{|\alpha|\beta}.$$

, - .
, $M_{0,+}$, $M_{0,-}$ - . , :

$$\lambda(t,t_0) = \pm 2\arctan\sinh\big(\alpha\beta(t-t_0)\big) + \lambda_-(x,y), \qquad \Lambda(t,t_0) = \frac{\pm 2\beta}{\cosh\big(\alpha\beta(t-t_0)\big)}, \tag{6}$$

$t_0 \in \mathbb{R}$, .

2.4.

$\tau = \varepsilon t$ (4), (τ):

$$\begin{cases} \varepsilon\Lambda' = U\sin\lambda - V\cos\lambda, & \varepsilon\lambda' = \alpha\Lambda, \\ x' = -\big(2Fy + \dfrac{\partial U}{\partial y}\cos\lambda + \dfrac{\partial V}{\partial y}\sin\lambda\big), & y' = 2Fx + e_J G + \dfrac{\partial U}{\partial x}\cos\lambda + \dfrac{\partial V}{\partial x}\sin\lambda. \end{cases} \tag{7}$$

ε , "" :

$$\begin{cases} 0 = U\sin\lambda - V\cos\lambda, & 0 = \alpha\Lambda, \\ x' = -\big(2Fy + \dfrac{\partial U}{\partial y}\cos\lambda + \dfrac{\partial V}{\partial y}\sin\lambda\big), & y' = 2Fx + e_J G + \dfrac{\partial U}{\partial x}\cos\lambda + \dfrac{\partial V}{\partial x}\sin\lambda. \end{cases} \tag{8}$$

(8) , :

$$\Big\{ x' = -\big(2Fy \pm \frac{\partial\beta}{\partial y}\big), \qquad y' = 2Fx + e_J G \pm \frac{\partial\beta}{\partial x}. \tag{9}$$

( ) $M_{0,+}$, $M_{0,-}$.
:

$$H_S = F(x^2+y^2) + e_J G x \pm \beta(x,y),$$

, .

$$(x_0^{\pm}, 0),$$

x_0 :

$$x_0^{\pm} = -\frac{e_J(G \pm \mathrm{sign}U(x_0^{\pm},0)D)}{2(F \pm \mathrm{sign}U(x_0^{\pm},0)C)}.$$

0 ( ), 1 2 (,). $M_{0,\pm}$,

$$U(x_0^{\pm},0) < 0.$$

, $U(x_0^{\pm},0) < 0.$

, ,

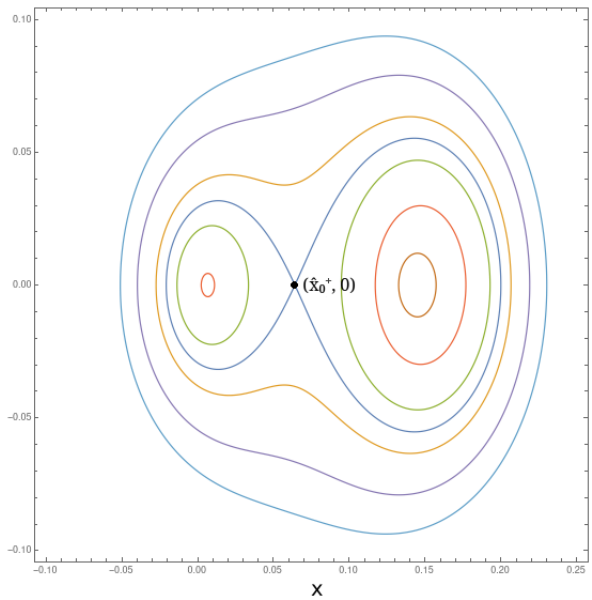
$$\zeta = \pm\sqrt{-\left(2F \pm 2C \pm \frac{2Cx_0^{\pm} + e_J D}{|U(x_0^{\pm},0)|}\right)\big(2F \mp 2C\big)}. \tag{10}$$

, , , .

, , . , , , .
, , , $\tau \to \pm\infty$. , :

$$H_S(x,y) = H_S(x_0^{\pm},0),$$

$$H_S(x_0^{\pm},0) = F\big((x_0^{\pm})^2\big) + e_J G x_0^{\pm} \pm \beta(x_0^{\pm},0).$$



. 5. H_S $M_{0,+}$. $M_{0,-}$

$x = x_0^{\pm} + h$, :

$$(F^2 - C^2)(h^2+y^2)^2 + 2(h^2+y^2)\big(U(x_0^{\pm},0) + h(2Cx_0^{\pm} + e_J D)\big)(\pm F - C) + \\ + y^2\big(4U(x_0^{\pm},0)C - (2Cx_0^{\pm} + e_J D)^2\big) = 0. \quad (11)$$

:

$$X = \frac{h}{h^2+y^2}, \quad Y = \frac{y}{h^2+y^2}.$$

(11) :

$$(F^2 - C^2) + AX^2 + 2BX + (A+\delta)Y^2 = 0,$$

$$A = 2U(x_0^{\pm},0)(\pm F - C), \quad B = (2Cx_0^{\pm} + e_J D)(\pm F - C), \quad \delta = -(2Cx_0^{\pm} + e_J D)^2.$$

$$\xi = \sigma_1\Big(X + \frac{B}{A}\Big), \quad \eta = \sigma_2 Y,$$

$$\sigma_1 = \sqrt{\frac{-A}{F^2 - C^2 - B^2/A}}, \quad \sigma_2 = \sqrt{\frac{A+\delta}{F^2 - C^2 - B^2/A}}.$$

$$0 < 2U(x_0^{\pm})(\pm F - C) < (2Cx_0^{\pm} + e_J D)^2.$$

$\sigma_2, \sigma_1 \in \mathbb{R}$.

$(x_0^{\pm}, 0)$

$$\xi^2 - \eta^2 = 1, \tag{12}$$

.. . (x, y) кнь (. . 5).

:

$$\xi = \cosh f, \quad \eta = \sinh f, \quad f \in \mathbb{R}.$$

f τ. $(x, y) \mapsto (X, Y) \mapsto (\xi, \eta)$,

$$f' = \frac{A+\delta}{\sigma_2} \frac{P(\cosh f)}{S(\cosh f)}, \tag{13}$$

$$P(s) = s^2 \left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2} \right) + s \frac{-2B}{\sigma_1 A} + \left(\frac{B^2}{A^2} - \frac{1}{\sigma_2^2} \right), \quad S(s) = \pm U(x_0^{\pm}, 0) P(s) \pm \left(\frac{s}{\sigma_1} - \frac{B}{A} \right) (2Cx_0^{\pm} + e_J D) + F.$$

, f:

$$\tau - \tau_0 = \frac{\sigma_2}{A+\delta} \int_{f_0}^{f} \frac{S(\cosh f)}{P(\cosh f)} df. \tag{14}$$

, $M_{0,+}$ $M_{0,-}$ (51) () $(x_0^{\pm}, 0)$, :

$$x(\tau, \tau_0) = x_0^{\pm} + \frac{\frac{\cosh(f(\tau-\tau_0))}{\sigma_1} - \frac{B}{A}}{P(\cosh(f(\tau-\tau_0)))}, \quad y(\tau, \tau_0) = \frac{\sinh(f(\tau-\tau_0))}{\sigma_2 P(\cosh(f(\tau-\tau_0)))}, \quad \tau_0 \in \mathbb{R}.$$

$f(\tau)$ (14).

2.5.

, $(\hat{x}_0^+, 0, \lambda_+(\hat{x}_0^+, 0), 0)$, $(\hat{x}_0^-, 0, \lambda_-(\hat{x}_0^-, 0), 0)$ (..), . ( ):

$$\pm \frac{\sqrt{q \pm \sqrt{p}}}{\sqrt{2}},$$

(M_0)

$$p = \left(4\varepsilon^2 (F^2 - C^2) \mp |\alpha| |U(\hat{x}_0^+, 0)| \right)^2 + 8\alpha\varepsilon^2 (F \mp C) V_y(\hat{x}_0^+, 0),$$
$$q = -4\varepsilon^2 (F^2 - C^2) \mp |\alpha| |U(\hat{x}_0^+, 0)|.$$

:

- $(\hat{x}_0^-, 0, \lambda_-(\hat{x}_0^-, 0), 0) \in M_{0,-}$ – - (, ,)
- $(\hat{x}_0^+, 0, \lambda_+(\hat{x}_0^+, 0), 0) \in M_{0,+}$ – - $(\pm i\omega, \pm\xi)$.

$$\omega = \frac{\sqrt{\sqrt{p} - q}}{\sqrt{2}} = \sqrt{|\alpha U(\hat{x}_0^+, 0)|} + \mathcal{O}(\varepsilon^2),$$
$$\xi = \frac{\sqrt{\sqrt{p} + q}}{\sqrt{2}} = \mathcal{O}(\varepsilon),$$

3.

3.1.

, . , [?].

:

$$\begin{cases} \frac{dX}{dt} = JD_X H_1(X,I) + \varepsilon g^X(X,I), \\ \frac{dI}{dt} = \varepsilon g^I(X,I), \end{cases} \tag{15}$$

$X \in \mathbb{R}^2$, $I \in \mathbb{R}^m$, D_X, D_I - , J - .
[?], $V \subset M_0$ - M_0 (15), V_ε - ε- V, . $\Gamma = W^s(V) \cap W^u(V) \setminus V$.
$W^s(V_\varepsilon)$ $W^u(V_\varepsilon)$ $p \in \Gamma$:

$$d^I(p,t_0,\varepsilon) = \varepsilon \frac{M^I(p,t_0)}{||D_X H(X_0^I, I)||} + O(\varepsilon^2), \tag{16}$$

M^I - , :

$$M^I(p,t_0) = \int_{-\infty}^{\infty} \Big(\langle D_X H_1, g^X\rangle + \langle D_I H_1, g^I\rangle\Big)\big(X_0^I(t-t_0,I),I\big)dt-$$
$$-\langle D_I H_1\big(v(I),I\big), \int_{-\infty}^{\infty} g^I\big(X_0^I(t-t_0,I),I\big)dt\rangle.$$

$X = v(I)$ - V, $\langle\cdot,\cdot\rangle$ - $\mathbb{R}^2$.
$W^s(V_\varepsilon)$ $W^u(V_\varepsilon)$ $\varepsilon > 0$:

$$M^I = 0, \quad \nabla M^I \neq 0. \tag{17}$$

, , ε d^I .
.

3.2.

, M_0 3:1 $M_{0,+}$, $M_{0,-}$ $M_{0,b}^{\pm}$, $M_{0,+}$, $M_{0,-}$.
, $M_{0,-}$, (??).
, (8) $M_{0,-}$ $(x(t), y(t), 0, \lambda_-(x(t), y(t)), \Lambda = 0)$ xy .
(4) (15) :

$$X = \begin{pmatrix} \Lambda \\ z \end{pmatrix}, \quad I = \begin{pmatrix} x \\ y \end{pmatrix},$$

$z = \lambda - \lambda_-(x,y)$.
$M_{0,-}$:

$$\tilde{M}_{0,-} = \{(\Lambda, z, x, y) : \Lambda = 0,\ z = 0\},$$

:

$$z_{sep}(t,t_0) = \pm 2 \operatorname{arctan\,sh}(\alpha\beta(t-t_0)), \quad \Lambda_{sep}(t,t_0) = \frac{2\beta}{\operatorname{ch}(\alpha\beta(t-t_0))}.$$

:

$$D_X = \begin{pmatrix} \frac{\partial}{\partial\Lambda} \\ \frac{\partial}{\partial z} \end{pmatrix}, \quad D_I = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{pmatrix},$$

$$H_1 = \alpha\frac{\Lambda^2}{2} + \beta(x,y)(1-\cos z), \quad \beta(x,y) = \sqrt{U(x,y)^2 + V(x,y)^2},$$

$$g^I = \begin{pmatrix} -2Fy + \frac{\partial\beta}{\partial y}\cos z + \beta(x,y)\frac{\partial}{\partial y}\arctan\left(\frac{V}{U}\right)\sin z \\ 2Fx + e_J G - \frac{\partial\beta}{\partial x}\cos z - \beta(x,y)\frac{\partial}{\partial x}\arctan\left(\frac{V}{U}\right)\sin z \end{pmatrix}, \quad g^X = \frac{-1}{\varepsilon}\begin{pmatrix} 0 \\ \frac{d}{dt}\arctan\left(\frac{V}{U}\right) \end{pmatrix}.$$

3.3.

$V = \tilde{M}_{0,-}$:

$$M^I(x,y) = \frac{4}{\alpha\beta^2}\left(U\left((2Fx + e_J G)\frac{\partial U}{\partial y} - 2Fy\frac{\partial U}{\partial x}\right) + V\left((2Fx + e_J G)\frac{\partial V}{\partial y} - 2Fy\frac{\partial V}{\partial x}\right)\right). \tag{18}$$

:

$$0 = y\left((2x+\tilde{\alpha})^2 + y^2 - (2x+\tilde{\alpha})\frac{4FK}{C(e_J G - F\tilde{\alpha})} - \frac{K}{C}\right), \tag{19}$$

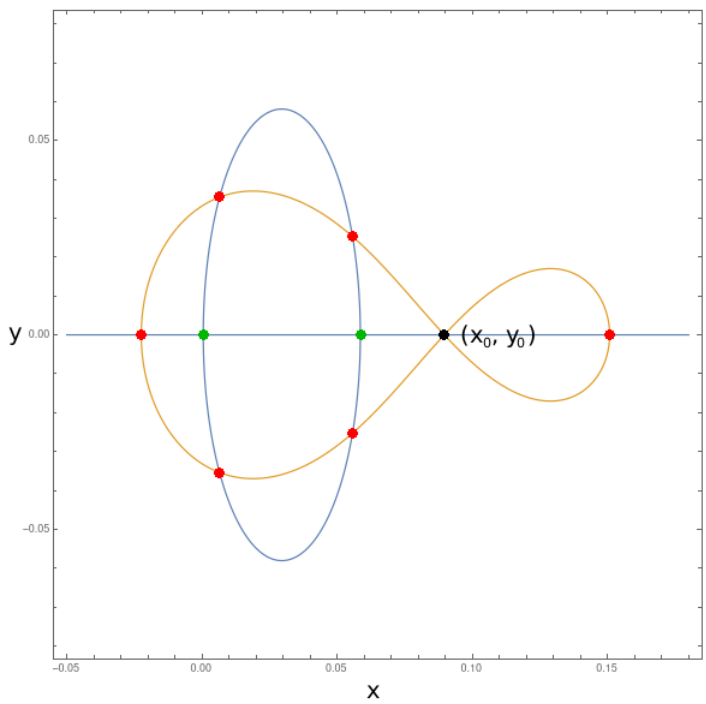
$y = 0$ (. 6).

, , .

:

1) $M_{0,-}$, , $\varepsilon > 0$ ( ).

2) $\left(x_0^-, y_0^-, 0, \lambda_-(x_0^-, y_0^-)\right)$ $\varepsilon > 0$, . $\left(x_0^-, y_0^-, 0, \lambda_-(x_0^-, y_0^-)\right)$, (. 6).



. 6. () (). – M^I

3.4.

, μ (1), . , .
$(x, y, \Lambda, \lambda) = \left(x_0^-, y_0^-, 0, \lambda_-(x_0^-, y_0^-)\right) \in M_{0,-}$, , , $t \to \pm\infty$. (1).

4.

.
$(\hat{x}_0^+, 0, \lambda_+(\hat{x}_0^+, 0), 0) \in M_{0,+}$, , (4). , - :

$$(\pm i\omega, \pm\xi),$$

$$\omega = \frac{\sqrt{\sqrt{p}-q}}{\sqrt{2}} = \sqrt{|\alpha U(\hat{x}_0^+, 0)|} + \mathcal{O}(\varepsilon^2),$$
$$\xi = \frac{\sqrt{\sqrt{p}+q}}{\sqrt{2}} = \mathcal{O}(\varepsilon),$$

$$p = (4\varepsilon^2(C-F)(C+F) + \alpha U(\hat{x}_0^+, 0))^2 + 8\alpha\varepsilon^2(F-C)V_y(\hat{x}_0^+, 0) >, 0$$
$$q = 4\varepsilon^2(C-F)(C+F) - \alpha U(\hat{x}_0^+, 0) < 0.$$

- . , , . ε, , , .

4.1.

. , , : , .
- 0, :

$$\Lambda \to \Lambda, \qquad \lambda \to \lambda - \lambda_+(\hat{x}_0, 0) = \lambda - \pi, \qquad x \to x - \hat{x}_0, \qquad y \to y.$$

:

$$\begin{cases} \dot{\Lambda} = -U\sin\lambda + V\cos\lambda \equiv g^{\Lambda}(\Lambda, \lambda, x, y), , & \dot{\lambda} = \alpha\Lambda \equiv g^{\lambda}(\Lambda, \lambda, x, y), \\ \dot{x} = -\varepsilon\big(2Fy - \dfrac{\partial U}{\partial y}\cos\lambda - \dfrac{\partial V}{\partial y}\sin\lambda\big) \equiv g^{x}(\Lambda, \lambda, x, y), & \dot{y} = \varepsilon\big(2F(x+\hat{x}_0) + e_J G - \dfrac{\partial U}{\partial x}\cos\lambda - \dfrac{\partial V}{\partial x}\sin\lambda\big) \equiv g^{y}(\Lambda, \lambda, x, y). \end{cases} \tag{20}$$

$$U(x,y) = U_0 + x(2C\hat{x}_0 + e_J D) + C(x^2 - y^2), \qquad V(x,y) = 2Cxy + y(2C\hat{x}_0 + e_J D),$$
$$U_0 = \text{const}.$$

, (20)

$$\Omega = d\Lambda \wedge d\lambda + \varepsilon^{-1} dx \wedge dy,$$

:

$$H = \frac{\alpha\Lambda^2}{2} - U(x,y)\cos\lambda - V(x,y)\sin\lambda + F\left((x+\hat{x}_0)^2 + y^2\right) + e_J G x. \tag{21}$$

4.1.1.

(20), $W^{s,u}(0)$. (20) :

- $W^s(0)$:
$$\Big\{\Lambda^s(+\infty) = 0, \quad \lambda^s(+\infty) = 0, \quad x^s(+\infty) = 0, \quad y^s(+\infty) = 0,$$
- $W^u(0)$:
$$\Big\{\Lambda^u(-\infty) = 0, \quad \lambda^u(-\infty) = 0, \quad x^u(-\infty) = 0, \quad y^u(-\infty) = 0.$$

(20) :

$$\Big\{\Lambda^{s,u}(\tau)=\sum_{k=0}^{\infty}\varepsilon^k\Lambda_k^{s,u}(\tau),\quad \lambda^{s,u}(\tau)=\sum_{k=0}^{\infty}\varepsilon^k\lambda_k^{s,u}(\tau),\quad x^{s,u}(\tau)=\sum_{k=0}^{\infty}\varepsilon^k x_k^{s,u}(\tau),\quad y^{s,u}(\tau)=\sum_{k=0}^{\infty}\varepsilon^k y_k^{s,u}(\tau), \tag{22}$$

, $\tau=\varepsilon t.$ ε, , , 2 , $(\lambda_k,\Lambda_k), k\geq 1$:

$$\begin{cases}\Lambda_k(\tau)=\frac{1}{\alpha}\lambda'_{k-1},\\ \lambda_k(\tau)=\frac{1}{\beta}\Big(\alpha^{-1}\lambda''_{k-2}-G_k^{\Lambda}+x_k\big((2Cx_0+2C\hat{x}_0+e_JD)\sin\lambda_0-2Cy_0\cos\lambda_0\big)+\\ \qquad +y_k\big(-2Cy_0\sin\lambda_0-(2Cx_0+2C\hat{x}_0+e_JD)\cos\lambda_0\big)\Big),\\ \lambda_{-1}=0,\end{cases} \tag{23}$$

$(x_k,y_k), k\geq 1$ - :

$$\frac{d}{d\tau}\begin{pmatrix}x_k\\ y_k\end{pmatrix}=\begin{pmatrix}a(\tau) & b(\tau)\\ c(\tau) & -a(\tau)\end{pmatrix}\begin{pmatrix}x_k\\ y_k\end{pmatrix}+\begin{pmatrix}\tilde{G}_k^x(\tau)\\ \tilde{G}_k^y(\tau)\end{pmatrix}\equiv\mathcal{A}(\tau)\begin{pmatrix}x_k\\ y_k\end{pmatrix}+\begin{pmatrix}\tilde{G}_k^x(\tau)\\ \tilde{G}_k^y(\tau)\end{pmatrix}, \tag{24}$$

:

$$\tilde{G}_k^x=G_k^x-\beta^{-1}(\alpha^{-1}\lambda''_{k-2}-G_k^{\Lambda})\big(-2Cy_0\sin\lambda_0-(2Cx_0+2C\hat{x}_0+e_JD)\cos\lambda_0\big),$$
$$\tilde{G}_k^y=G_k^y-\beta^{-1}(\alpha^{-1}\lambda''_{k-2}-G_k^{\Lambda})\big(-(2Cx_0+2C\hat{x}_0+e_JD)\sin\lambda_0+2Cy_0\cos\lambda_0\big).$$

a,b,c - $x_0(\tau),y_0(\tau),\lambda_0(\tau),\Lambda_0(\tau)$:

$$a(\tau)=\frac{(e_J^2D^2-4CEe_J^2)\sin\lambda_0\cos\lambda_0}{\beta(x_0,y_0)},$$

$$b(\tau)=-2F-2C\cos\lambda_0-\frac{1}{\beta(x_0,y_0)}\big((2Cy_0)\sin\lambda_0+(2Cx_0+2C\hat{x}_0+e_JD)\cos\lambda_0\big)^2,$$

$$c(\tau)=2F-2C\cos\lambda_0+\frac{1}{\beta(x_0,y_0)}\big((2Cx_0+2C\hat{x}_0+e_JD)\sin\lambda_0-(2Cy_0)\cos\lambda_0\big)^2.$$

1. 1) $G_k^{\rho}, \rho\in\{\lambda,\Lambda,x,y\}$, (24), :

$$G_k^{\rho}=\sum_{j=2}^{k}\frac{\mathcal{L}_k^{(j)}g^{\rho}}{j!},\quad k\geq 2,$$

g^{ρ} - , $\rho\in\{\lambda,\Lambda,x,y\}$,

$$\mathcal{L}_k^{(j)}=\sum_{k_1+\ldots+k_j=k}\mathfrak{D}_{k_1}\cdot\ldots\cdot\mathfrak{D}_{k_j},\qquad \mathfrak{D}_k=\left(\begin{pmatrix}\lambda_k\\ \Lambda_k\\ x_k\\ y_k\end{pmatrix},\begin{pmatrix}\frac{\partial}{\partial\lambda}\big|_{\lambda_0,\Lambda_0,x_0,y_0}\\ \frac{\partial}{\partial\Lambda}\big|_{\lambda_0,\Lambda_0,x_0,y_0}\\ \frac{\partial}{\partial x}\big|_{\lambda_0,\Lambda_0,x_0,y_0}\\ \frac{\partial}{\partial y}\big|_{\lambda_0,\Lambda_0,x_0,y_0}\end{pmatrix}\right),$$

$(\cdot,\cdot)$ – $\mathbb{R}^4$.

2) :

$$\mathcal{F}_s=\Big\{f\colon\mathbb{R}\to\mathbb{R}\ \Big|\ \exists C>0:\ \forall x\in\mathbb{R}\Rightarrow\ |f(x)|\leq Ce^{-s|x|}\Big\}, s>0$$

$$\mathcal{F}_{-s}=\Big\{f\colon\mathbb{R}\to\mathbb{R}\ \Big|\ \exists C>0:\ \forall x\in\mathbb{R}\Rightarrow\ |f(x)|\geq Ce^{s|x|}\Big\}, s>0$$

$\lambda_j,\Lambda_j,x_j,y_j\in\mathcal{F}_s\ \forall\ 0\leq j\leq k,\ \ g^{\rho}\in C_b^{\infty}(\mathbb{R}^4)$ ($C_b^{\infty}(\mathbb{R}^4)$,).
$G_{k+1}^{\rho}\in\mathcal{F}_{2s}$

3)

$$G_k^\lambda = 0 \quad \forall k \geq 1.$$

$\varepsilon = 0.$, $W^{s,u}(0)$, :

$$\begin{cases} \Lambda_0(\tau) = 0, \quad \lambda_0(\tau) = \lambda_-(x_{sep}(\tau), y_{sep}(\tau)), \\ x_0(\tau) = x_{sep}(\tau) - \hat{x}_0, \quad y_0(\tau) = y_{sep}(\tau). \end{cases}$$

(x_1, y_1) :

$$\frac{d}{d\tau}\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \mathcal{A}(\tau)\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}. \tag{25}$$

. ( ). $\tilde{G}_1^x$ $\tilde{G}_1^y$ - .
(25) :

$$\left\{ x_1^{s,u}(\tau) = \frac{dx_0}{d\tau}, \qquad y_1^{s,u}(\tau) = \frac{dy_0}{d\tau}, \qquad \boldsymbol{u}_1(\tau) \equiv \begin{pmatrix} x_1(\tau) \\ y_1(\tau) \end{pmatrix} \right. \tag{26}$$

(25) ( ) $\tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau) \equiv (\tilde{x}_1(\tau), \tilde{y}_1(\tau))$. .. $\mathcal{A}$ 0, , -, , , :

$$W(\tau) = x_1(\tau)\tilde{y}_1(\tau) - y_1(\tau)\tilde{x}_1(\tau) = 1$$

(x_1, y_1) $(\tilde{x}_1, \tilde{y}_1)$.
$|\tau|$ $\mathcal{A}(\tau)$:

$$\mathcal{A}(\tau) = \underbrace{\begin{pmatrix} 0 & -2F - 2C + \frac{(2C\hat{x}_0 + e_J D)^2}{U_0} \\ 2F - 2C & 0 \end{pmatrix}}_{\mathcal{A}_0} + \mathcal{O}(e^{-s_0|\tau|}) \equiv \begin{pmatrix} 0 & b_0 \\ c_0 & 0 \end{pmatrix} + \mathcal{O}(e^{-s_0|\tau|}), \quad s_0 \equiv \sqrt{b_0 c_0} \in \mathbb{R} \tag{27}$$

:

$$\boldsymbol{u}_1(\tau) = \text{const}\begin{pmatrix} b_0\sqrt{c_0} \\ c_0\sqrt{b_0} \end{pmatrix} e^{-s_0\tau} + \mathcal{O}(e^{-2s_0\tau}), \qquad \tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau) = \text{const}\begin{pmatrix} b_0\sqrt{c_0} \\ -c_0\sqrt{b_0} \end{pmatrix} e^{s_0\tau} + \mathcal{O}(1),$$

τ :

$$\boldsymbol{u}_1(\tau) = \text{const}\begin{pmatrix} b_0\sqrt{c_0} \\ -c_0\sqrt{b_0} \end{pmatrix} e^{s_0\tau} + \mathcal{O}(e^{2s_0\tau}), \qquad \tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau) = \text{const}\begin{pmatrix} b_0\sqrt{c_0} \\ c_0\sqrt{b_0} \end{pmatrix} e^{-s_0\tau} + \mathcal{O}(1),$$

τ.
:

$$\boldsymbol{u}_{\mathbf{1}}(\tau) \in \mathcal{F}_{s_0}, \qquad \tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau) \in \mathcal{F}_{-s_0}.$$

, (x_1, y_1) $W^s(0)$, $W^u(0)$, s, u .
$\mathcal{A}$, . , , :

$$\begin{cases} \boldsymbol{u}_k^u = \boldsymbol{u}_1(\tau)\displaystyle\int_{\tau_k^u}^{\tau}\left(\tilde{y}_1(\tau)\tilde{G}_k^x(\tau) - \tilde{x}_1(\tau)\tilde{G}_k^y(\tau)\right)d\tau + \tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau)\int_{-\infty}^{\tau}\left(x_1(\tau)\tilde{G}_k^y(\tau) - y_1(\tau)\tilde{G}_k^x(\tau)\right)d\tau, \\ \boldsymbol{u}_k^s = \boldsymbol{u}_1(\tau)\displaystyle\int_{\tau_k^s}^{\tau}\left(\tilde{y}_1(\tau)\tilde{G}_k^x(\tau) - \tilde{x}_1(\tau)\tilde{G}_k^y(\tau)\right)d\tau - \tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau)\int_{\tau}^{+\infty}\left(x_1(\tau)\tilde{G}_k^y(\tau) - y_1(\tau)\tilde{G}_k^x(\tau)\right)d\tau. \end{cases} \tag{28}$$

, (28) $\tau \to \pm\infty$. , (28) τ .
$\mathbf{u}_1(\tau)$ $\tau \to \pm\infty$, $\mathbf{u}_k^u$ $\mathbf{u}_k^s$ $\tau_k^u, \tau_k^s \in \mathbb{R}$.

$\tilde{\mathbf{u}}_1(\tau)$ $\tau \to \pm\infty$. $\mathbf{u}_k^u$ $\mathbf{u}_k^s$, (28), . $(x_1\tilde{G}_k^y - y_1\tilde{G}_k^x) \in \mathcal{F}_{3s_0}$, , , $\tilde{\mathbf{u}}_1$.

$$\Lambda^s(\tau) = \sum_{k=0}^{\infty} \varepsilon^k \Lambda_k(\tau), \quad \lambda^s(\tau) = \sum_{k=0}^{\infty} \varepsilon^k \lambda_k(\tau), \quad x^s(\tau) = \sum_{k=0}^{\infty} \varepsilon^k x_k(\tau), \quad y^s(\tau) = \sum_{k=0}^{\infty} \varepsilon^k y_k(\tau). \tag{29}$$

(x_1, y_1) (24) $(x_1(\tau), y_1(\tau)) = \text{const} \cdot \boldsymbol{u}_1$, τ_k^s .
- , $W^s(0)$. (20), $W^s(0)$. (29).
$W^u(0)$, , $+\infty$.
:

$$|\boldsymbol{u}_k^u(\tau) - \boldsymbol{u}_k^s(\tau)| = \Bigg| \boldsymbol{u}_1(\tau) \int_{\tau_k^u}^{\tau_k^s} \left(\tilde{y}_1(\tau)\tilde{G}_k^x(\tau) - \tilde{x}_1(\tau)\tilde{G}_k^y(\tau) \right) d\tau + \\ + \tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau) \int_{-\infty}^{+\infty} \left(x_1(\tau)\tilde{G}_k^y(\tau) - y_1(\tau)\tilde{G}_k^x(\tau) \right) d\tau \Bigg| \tag{30}$$

τ . $W^s(0)$ $W^u(0)$ $\boldsymbol{u}_k^s$ $\boldsymbol{u}_k^u$ τ.
, $|\boldsymbol{u}_k^u(\tau_1) - \boldsymbol{u}_k^s(\tau_2)| = 0$ τ_1, τ_2. , , $\tau_1 = \tau_2 = \tau$.

1. , $f(\tau)$, $f(\tau)$, τ.

, $\lambda_j, \Lambda_j, x_j, y_j$, $j \le k-1$ τ . (20) , $G_k^x(\tau), \tilde{G}_k^x(\tau), G_k^\Lambda(\tau), G_k^y(\tau)$. $G_k^x(\tau), \tilde{G}_k^x(\tau), G_k^\Lambda(\tau)$, $G_k^y(\tau), \tilde{G}_k^y(\tau)$ $G_k^x(\tau)$ ( ).
$G_2^x(\tau), \tilde{G}_2^x(\tau), G_2^\Lambda$ - , $G_2^y(\tau), \tilde{G}_2^y(\tau)$ - .
, , :

$$\int_{-\infty}^{+\infty} \left(x_1(\tau)\tilde{G}_2^y(\tau) - y_1(\tau)\tilde{G}_2^x(\tau) \right) d\tau = 0.$$

4.1.2.

, 0.
, :

$$\boldsymbol{u}_1^0(\tau) \equiv \begin{pmatrix} x_1^0(\tau) \\ y_1^0(\tau) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

$\tilde{G}_2^\rho, \rho \in \{\Lambda, \lambda, x, y\}$.
$\tau_2^s = \tau_2^u = 0$:

$$\begin{cases} \boldsymbol{u}_2^u = \boldsymbol{u}_1(\tau) \int_0^\tau \left(\tilde{y}_1(\tau)\tilde{G}_2^x(\tau) - \tilde{x}_1(\tau)\tilde{G}_2^y(\tau) \right) d\tau + \tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau) \int_{-\infty}^{\tau} \left(x_1(\tau)\tilde{G}_2^y(\tau) - y_1(\tau)\tilde{G}_2^x(\tau) \right) d\tau, \\ \boldsymbol{u}_2^s = \boldsymbol{u}_1(\tau) \int_0^\tau \left(\tilde{y}_1(\tau)\tilde{G}_2^x(\tau) - \tilde{x}_1(\tau)\tilde{G}_2^y(\tau) \right) d\tau - \tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau) \int_{\tau}^{+\infty} \left(x_1(\tau)\tilde{G}_2^y(\tau) - y_1(\tau)\tilde{G}_2^x(\tau) \right) d\tau, \end{cases}$$

, (??)

$$\boldsymbol{u}_2^s(\tau) = \boldsymbol{u}_2^u(\tau) \equiv \boldsymbol{u}_2^0(\tau).$$

$\tilde{G}_k^\rho, \rho \in \{\Lambda, \lambda, x, y\}$. 3 . :

$$\boldsymbol{u}_3^0(\tau) \equiv \begin{pmatrix} x_3^0(\tau) \\ y_3^0(\tau) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \qquad (\lambda_3^0(\tau), \Lambda_3^0(\tau)) = (0, 0).$$

2. $k \geq 0$

$$G^{\rho}_{2k+1} = 0, \rho \in \{\Lambda, \lambda, x, y\}, \qquad G^{x}_{2k+2}, y_{2k+2} - , \qquad G^{y}_{2k+2}, x_{2k+2} - .$$

:

$$(\lambda^0_{2k+1}, \Lambda^0_{2k+1}, x^0_{2k+1}, y^0_{2k+1}) = (0, 0, 0, 0).$$

:

$$\begin{cases} \lambda^0(\tau) = (\lambda_0(\tau) - \pi) + \sum_{k=1}^{+\infty} \varepsilon^{2k} \lambda^0_{2k}(\tau), \\ \Lambda^0(\tau) = \frac{1}{\alpha} \sum_{k=1}^{+\infty} \varepsilon^{2k} (\lambda^0_{2k-1})'(\tau), \\ x^0(\tau) = x_0(\tau) + \sum_{k=1}^{+\infty} \varepsilon^{2k} x^0_{2k}(\tau), \\ y^0(\tau) = y_0(\tau) + \sum_{k=1}^{+\infty} \varepsilon^{2k} y^0_{2k}(\tau), \end{cases}$$

$$\begin{cases} x^0_{2k}(\tau) = x_1(\tau) \int_0^{\tau} \left(\tilde{y}_1(\tau) \tilde{G}^x_{2k}(\tau) - \tilde{x}_1(\tau) \tilde{G}^y_{2k}(\tau) \right) d\tau + \tilde{x}_1(\tau) \int_{-\infty}^{\tau} \left(x_1(\tau) \tilde{G}^y_{2k}(\tau) - y_1(\tau) \tilde{G}^x_{2k}(\tau) \right) d\tau, \\ y^0_{2k}(\tau) = y_1(\tau) \int_0^{\tau} \left(\tilde{y}_1(\tau) \tilde{G}^x_{2k}(\tau) - \tilde{x}_1(\tau) \tilde{G}^y_{2k}(\tau) \right) d\tau + \tilde{y}_1(\tau) \int_{-\infty}^{\tau} \left(x_1(\tau) \tilde{G}^y_{2k}(\tau) - y_1(\tau) \tilde{G}^x_{2k}(\tau) \right) d\tau, \end{cases}$$

$$\tilde{G}^x_{2k} = \frac{1}{\alpha\beta} (\lambda^0_{2k-2})''(\tau) \left(-2Cy_0 \sin\lambda_0 - (2Cx_0 + 2C\hat{x}_0 + e_J D) \cos\lambda_0 \right),$$

$$\tilde{G}^y_{2k} = \frac{1}{\alpha\beta} (\lambda^0_{2k-2})''(\tau) \left(-(2Cx_0 + 2C\hat{x}_0 + e_J D) \sin\lambda_0 + 2Cy_0 \cos\lambda_0 \right).$$

1. (20):

- $(\Lambda^s(\tau), \lambda^s(\tau), x^s(\tau), y^s(\tau)) \in W^s(0)$,
- $(\Lambda^u(\tau), \lambda^u(\tau), x^u(\tau), y^u(\tau)) \in W^u(0)$,

  ε $(\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau))$. $\exists$ $c > 0$:

$$(\Lambda^s(\tau), \lambda^s(\tau), x^s(\tau), y^s(\tau)) = (\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau)) + \mathcal{O}(e^{-\frac{c}{\varepsilon}}),$$

$$(\Lambda^u(\tau), \lambda^u(\tau), x^u(\tau), y^u(\tau)) = (\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau)) + \mathcal{O}(e^{-\frac{c}{\varepsilon}}).$$

:

$$\int_0^{\tau} \left(\tilde{y}_1(\tau) \tilde{G}^x_{2k}(\tau) - \tilde{x}_1(\tau) \tilde{G}^y_{2k}(\tau) \right) d\tau,$$

$\forall k \in \mathbb{N}$, .. $\mathcal{F}_{s_0}$.
(2) 0:

$$\int_{-\infty}^{+\infty} \left(x_1(\tau) \tilde{G}^y_{2k}(\tau) - y_1(\tau) \tilde{G}^x_{2k}(\tau) \right) d\tau, \quad k \in \mathbb{N}$$

, $\mathcal{F}_{3s_0}$,

$$\int_{-\infty}^{\tau} \left(x_1(\tau) \tilde{G}^y_{2k}(\tau) - y_1(\tau) \tilde{G}^x_{2k}(\tau) \right) d\tau$$

$\mathcal{F}_{2s_0}$. $\boldsymbol{\tilde{u}_1}$.
, $(\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau))$ $\tau \to \pm\infty$.
, $(g^{\Lambda}, g^{\lambda}, g^x, g^y)$ (20) (Λ, λ, x, y) $\varepsilon > 0$. , $(g^{\Lambda}, g^{\lambda}, g^x, g^y)$ (Λ, λ, x, y) . .
, $W^s(0)$ $W^u(0)$ $(\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau))$. .
■

1. - (20) .

4.2.

[?], - . [?] [?, ?]. .

4.2.1.

(20) . :

$$A = \begin{pmatrix} 0 & -U_0 & 0 & (2C\hat{x}_0 + e_J D) \\ \alpha & 0 & 0 & 0 \\ 0 & \varepsilon(2C\hat{x}_0 + e_J D) & 0 & -\varepsilon(2F + 2C) \\ 0 & 0 & \varepsilon(2F - 2C) & 0 \end{pmatrix}.$$

A $\pm i\omega, \pm\xi$ $(z, \eta, a, b)^T = R^{-1}(\Lambda, \lambda, x, y)^T$, $R -$. :

$$\frac{d}{dt}(z, \eta, a, b)^T = \mathrm{diag}(i\omega, -i\omega, \xi, -\xi)(z, \eta, a, b)^T + \boldsymbol{\mathcal{G}}(z, \eta, a, b), \tag{31}$$

$\boldsymbol{\mathcal{G}}$.

$\phi(t), \psi(t)$, , [?]. :

$$\varphi(t) = \varphi_0 e^{i\theta(\varphi_0,\psi_0)t}, \qquad \psi(t) = \varphi_0 e^{-i\theta(\varphi_0,\psi_0)t}.$$

$\theta(\varphi_0, \psi_0)$ φ_0, ψ_0:

$$\theta(\varphi_0, \psi_0) = \omega + \sum_{k=1}^{+\infty} \theta_k \cdot (\varphi_0\psi_0)^k.$$

:

$$\begin{cases} z(\varphi, \psi) = \varphi + \sum_{k,j=1}^{\infty} \{z\}_{k,j} \varphi^k \psi^j, & \eta(\varphi, \psi) = \psi + \sum_{k,j=1}^{\infty} \{\eta\}_{k,j} \varphi^k \psi^j, \\ a(\varphi, \psi) = \sum_{k,j=1}^{\infty} \{a\}_{k,j} \varphi^k \psi^j, & b(\varphi, \psi) = \sum_{k,j=1}^{\infty} \{b\}_{k,j} \varphi^k \psi^j. \end{cases} \tag{32}$$

, z $(\varphi\psi)^k\varphi$, η $(\varphi\psi)^k\psi$. φ, ψ [?].
(31) :

$$\begin{cases} ((p-q)i\omega - i\omega)\{z\}_{p,q} + \sum_{r=1}^{+\infty} (p-q)i\theta_r \{z\}_{p-r,q-r} = \{\boldsymbol{\mathcal{G}}_z(z, \eta, a, b)\}_{p,q}, \\ ((p-q)i\omega + i\omega)\{\eta\}_{p,q} + \sum_{r=1}^{+\infty} (p-q)i\theta_r \{\eta\}_{p-r,q-r} = \{\boldsymbol{\mathcal{G}}_\eta(z, \eta, a, b)\}_{p,q}, \\ ((p-q)i\omega - \xi)\{a\}_{p,q} + \sum_{r=1}^{+\infty} (p-q)i\theta_r \{a\}_{p-r,q-r} = \{\boldsymbol{\mathcal{G}}_a(z, \eta, a, b)\}_{p,q}, \\ ((p-q)i\omega + \xi)\{b\}_{p,q} + \sum_{r=1}^{+\infty} (p-q)i\theta_r \{b\}_{p-r,q-r} = \{\boldsymbol{\mathcal{G}}_b(z, \eta, a, b)\}_{p,q}. \end{cases} \tag{33}$$

$\{\boldsymbol{\mathcal{G}}_\rho(z, \eta, a, b)\}_{p,q}, \rho \in \{z, \eta, a, b\}$ $\varphi^p\psi^q$ $\boldsymbol{\mathcal{G}}$.
$p = q + 1, p + q > 1$ z , $\{z\}_{p,q} = 0$. $q = p + 1$ η , $\{\eta\}_{p,q} = 0$ ($(\varphi\psi)^k\varphi$ z $(\varphi\psi)^k\psi$ η). $\{z\}_{1,0} = \{\eta\}_{0,1} = 1$. ,

$$\begin{cases} \theta_p = \{\boldsymbol{\mathcal{G}}_z\}_{p,q} & p = q + 1 > 1, \\ \theta_q = -\{\boldsymbol{\mathcal{G}}_\eta\}_{p,q} & q = p + 1 > 1. \end{cases}$$

(Λ, λ, x, y), , :

$$\varphi_0 = \psi_0 \equiv h.$$

4.2.2.

.

2. , $f(x)$, - 0,

$$g(s) = \sum_{n=0}^{+\infty} g_n s^n, \quad s \geq 0,$$
$$g_n \geq 0 \quad \forall n \geq 0,$$

:

$f(x)$:

$$f(x) = \sum_{n=0}^{+\infty} f_n x^n.$$

$$\exists C > 0 : |f_n| \leq C \cdot g_n \quad \forall n \geq 0.$$

$$f \prec g.$$

$\varkappa$:

$$\varkappa = \max\left(\sup_{p,q\in\mathbb{R}} \left(\left| \frac{1}{(p-q)i\omega - \xi} \right| \right), \sup_{p,q\in\mathbb{R}} \left(\left| \frac{p-q}{(p-q)i\omega - \xi} \right| \right), \sup_{p,q\in\mathbb{R}} \left(\left| \frac{1}{(p-q)i\omega - i\omega} \right| \right), \sup_{p,q\in\mathbb{R}} \left(\left| \frac{p-q}{(p-q)i\omega - i\omega} \right| \right) \right) =$$
$$= \max(\frac{1}{|\xi|}, \frac{1}{|\omega|}) = \frac{1}{|\xi|} = O\left(\frac{1}{\varepsilon}\right) \qquad \varepsilon.$$

, (33), :

$$\left\{ |\{\rho\}_{p,q}| \leq \varkappa |\{\boldsymbol{\mathcal{G}}_\rho(z,\eta,a,b)\}_{p,q}| + \varkappa \textstyle\sum_{r=1}^{+\infty} |(p-q)\theta_r \{\rho\}_{p-r,q-r}|, \qquad \rho \in \{z,\eta,a,b\} \right. \tag{34}$$

[?] , $\boldsymbol{\mathcal{G}}$

$$\boldsymbol{\mathcal{G}} \prec \frac{1}{1-s}, \qquad s \equiv |z| + |\eta| + |a| + |b|.$$

$\boldsymbol{\mathcal{G}}$.

.

(Λ, λ, x, y) (z, η, a, b) R, s:

$$s = |z| + |\eta| + \varepsilon(|a| + |b|).$$

R :

$$\left\{ \Lambda \prec s, \qquad \lambda \prec s, \qquad x \prec \varepsilon s, \qquad y \prec \varepsilon s. \right. \tag{35}$$

(35) , 0 , (20):

$$\begin{cases} -U\sin\lambda + V\cos\lambda \prec (\varepsilon s + \varepsilon^2 s^2)e^s, & \alpha\Lambda \prec s, \\ -\varepsilon\big(2Fy - \frac{\partial U}{\partial y}\cos\lambda - \frac{\partial V}{\partial y}\sin\lambda\big) \prec \varepsilon(\varepsilon s + \varepsilon^2 s^2)e^s, & \varepsilon\big(2F(x+\hat{x}_0) + e_J G - \frac{\partial U}{\partial x}\cos\lambda - \frac{\partial V}{\partial x}\sin\lambda\big) \prec \varepsilon(\varepsilon s + \varepsilon^2 s^2)e^s. \end{cases}$$

$\boldsymbol{\mathcal{G}}$:

$$\begin{cases} \boldsymbol{\mathcal{G}}_{z,\eta} & \prec (\varepsilon(s + \varepsilon s^2)e^s - \varepsilon s), \\ \boldsymbol{\mathcal{G}}_{a,b} & \prec ((s + \varepsilon s^2)e^s - s). \end{cases} \tag{36}$$

, $-s$ $(s+\varepsilon s^2)e^s$ $(s+\varepsilon s^2)e^s - s$.
h, φ, ψ, :

$$||h|| = \sum_{p,q} |\{h\}_{p,q}| \varphi^p \psi^q.$$

:

$$z^* = z - \varphi, \qquad \eta^* = \eta - \psi, \qquad a^* = a, \qquad b^* = b, \qquad \theta^* = \theta - \omega.$$

(34) ε a b, $\varphi^p \psi^q$. :

$$(\varphi + \psi)||\theta^*|| + S \prec \varkappa\big(||\boldsymbol{\mathcal{G}}_z|| + ||\boldsymbol{\mathcal{G}}_\eta|| + \varepsilon(||\boldsymbol{\mathcal{G}}_a|| + ||\boldsymbol{\mathcal{G}}_b||) + ||\theta^*||S\big), \tag{37}$$

$S = ||z^*|| + ||\eta^*|| + \varepsilon(||a^*|| + ||b^*||)$.
S $||\theta^*||$ $\varphi = 0, \psi = 0.$ $\varphi = \psi > 0.$
:

$$2\varphi||\theta^*|| + S \prec \varkappa\left(\varepsilon(s + \varepsilon s^2)e^s - \varepsilon s + ||\theta^*||S\right),$$

$s \prec 2\varphi + S$.
$Y = 2||\theta^*|| + \frac{1}{\varphi}S$, . :

$$Y \prec \frac{\varkappa}{4}\left(Y^2 + \varepsilon\left((2+Y) + \varepsilon\varphi(2+Y)^2\right)e^{\varphi(2+Y)} - (2+Y)\right). \tag{38}$$

$$\hat{Y} = \frac{\varkappa}{4}\left(\hat{Y}^2 + \varepsilon\left((2+\hat{Y}) + \varepsilon\varphi(2+\hat{Y})^2\right)e^{\varphi(2+\hat{Y})} - (2+\hat{Y})\right),$$

$\hat{Y} = \sum_{k=1}^{\infty} \gamma_k \varphi^k$. , $R = \mathcal{O}(\varepsilon)$, φ.

4.3.

-.
:

$$\Gamma_h(t) = (\Lambda^*(h,t), \lambda^*(h,t), x^*(h,t), y^*(h,t)).$$

, $W^{s,u}(\Gamma_h)$, :

$$\begin{cases} \Lambda^{s,u}(h,t) = \Lambda^*(h,t) + z_\Lambda^{s,u}(h,t), & \lambda^{s,u}(h,t) = \lambda^*(h,t) + z_\lambda^{s,u}(h,t), \\ x^{s,u}(h,t) = x^*(h,t) + z_x^{s,u}(h,t), & y^{s,u}(h,t) = y^*(h,t) + z_y^{s,u}(h,t), \end{cases} \tag{39}$$

(39) (20). :

- $W^s(\Gamma_h)$:

$$\Big\{ z_\Lambda^s(k,+\infty) = 0, \qquad z_\lambda^s(k,+\infty) = 0, \qquad z_x^s(k,+\infty) = 0, \qquad z_y^s(k,+\infty) = 0.$$

- $W^u(\Gamma_h)$:

$$\Big\{ z_\Lambda^u(k,-\infty) = 0, \qquad z_\lambda^u(k,-\infty) = 0, \qquad z_x^u(k,-\infty) = 0, \qquad z_y^u(k,-\infty) = 0.$$

$\tilde{\tau} = \frac{\theta(h)}{\omega}\tau = (1 + \mathcal{O}(h^2))\tau$. , h h $\frac{\omega}{\varepsilon}$. :

$$H_{old} \to H = \frac{\omega}{\theta(h)} H_{old} = H_{old} + \mathcal{O}(h^2).$$

$\frac{\omega}{\varepsilon} = \mathcal{O}(1/\varepsilon)$ $\frac{\xi}{\varepsilon} = \mathcal{O}(1)$, ($\langle\rangle$).
(39) (21) $\boldsymbol{z}$. :

$$\langle H\rangle = \frac{\omega}{\theta(h)}\Bigg(H_{old}(\boldsymbol{z}) + \alpha\langle\Lambda^*\rangle z_\Lambda + 2F\langle x^*\rangle z_x - \langle\cos\lambda^* - 1\rangle\Big(U(z_x,z_y)\cos z_\lambda + V(z_x,z_y)\sin z_\lambda\Big) -$$
$$- \cos z_\lambda\Big((2C\hat{x}_0 + e_J D)\langle y^*\sin\lambda^*\rangle + z_x 2C\left(\langle x^*\cos\lambda^*\rangle + \langle y^*\sin\lambda^*\rangle\right) + \Big\langle\big(U(x^*,y^*) - U_0\big)\cos\lambda^*\Big\rangle\Big) -$$
$$- \sin z_\lambda\Big(z_y 2C\left(\langle x^*\cos\lambda^*\rangle + \langle y^*\sin\lambda^*\rangle\right)\Big)\Bigg) + \mathcal{O}(h^4), \quad (40)$$

, h^2:

$$\langle H\rangle_0 = H_{old}(\boldsymbol{z}) - h^2\theta_1 H_{old}(\boldsymbol{z}) + \alpha\langle\Lambda^*\rangle_0 z_\Lambda + 2F\langle x^*\rangle_0 z_x -$$
$$- \langle\cos\lambda^* - 1\rangle_0\Big(U(z_x,z_y)\cos z_\lambda + V(z_x,z_y)\sin z_\lambda\Big) -$$
$$- \cos z_\lambda\Big((2C\hat{x}_0 + e_J D)\langle y^*\sin\lambda^*\rangle_0 + z_x 2C\left(\langle x^*\cos\lambda^*\rangle_0 + \langle y^*\sin\lambda^*\rangle_0\right) + \Big\langle\big(U(x^*,y^*) - U_0\big)\cos\lambda^*\Big\rangle_0\Big) -$$
$$- \sin z_\lambda\Big(z_y 2C\left(\langle x^*\cos\lambda^*\rangle_0 + \langle y^*\sin\lambda^*\rangle_0\right)\Big). \quad (41)$$

$\langle\cdot\rangle_0$ h^2, .
h, (41) (40)

$$(\hat{\Lambda}_0^*(h), \hat{\lambda}_0^*(h), \hat{x}_0^*(h), \hat{y}_0^*(h)) \qquad (\hat{\Lambda}_0^{**}(h), \hat{\lambda}_0^{**}(h), \hat{x}_0^{**}(h), \hat{y}_0^{**}(h)),$$

, , $\mathcal{O}(h^2)$- .
, , [?, ?], ( ). , h , .
$\langle H\rangle_0$ 0. :

$$\langle H\rangle_0 = \frac{\alpha\omega z_\Lambda^2}{2\theta(h)} - \hat{U}(z_x,z_y)\cos z_\lambda - \hat{V}(z_x,z_y)\sin z_\lambda + \frac{F\omega}{\theta(h)}\left((z_x + \hat{x}_0 + \hat{x}_0^*(h))^2 + z_y^2\right) +$$
$$+ \frac{\omega}{\theta(h)}\left(e_J G + 2F\langle x^*\rangle_0 + 2F\hat{x}_0^*(h)\right) z_x, \quad (42)$$

$\tilde{U}$ $\tilde{V}$ U, V $\mathcal{O}(h^2)$:

$$\begin{cases} \hat{U}(x,y) = \underbrace{C\left(\langle\cos\lambda^*\rangle_0 - h^2\theta_1\right)}_{C+\mathcal{O}(h^2)}(x^2 - y^2) + \\ +x\underbrace{\left((2C\hat{x}_0 + e_J D)\left(\langle\cos\lambda^*\rangle_0 - h^2\theta_1\right) + 2C\left(\langle\cos\lambda^*\rangle_0 - h^2\theta_1\right)\hat{x}_0^*(h) + T\right)}_{(2C\hat{x}_0 + e_J D)+\mathcal{O}(h^2)} + \\ +\Big((2C\hat{x}_0 + e_J D)\langle y^*\sin\lambda^*\rangle_0 + \langle U(x^*,y^*)\cos\lambda^*\rangle_0 + T\hat{x}_0^*(h) + C\left(\langle\cos\lambda^*\rangle_0 - h^2\theta_1\right)(\hat{x}_0^*(h))^2 + \\ +(2C\hat{x}_0 + e_J D)\left(\langle\cos\lambda^*\rangle_0 - h^2\theta_1\right)x_0^*(h)\Big), \\ \\ \hat{V}(x,y) = 2C\langle\cos\lambda^*\rangle_0 xy + \\ +y\Big((2C\hat{x}_0 + e_J D)\left(\langle\cos\lambda^*\rangle_0 - h^2\theta_1\right) + 2C\left(\langle\cos\lambda^*\rangle_0 - h^2\theta_1\right)\hat{x}_0^*(h) + T\Big), \end{cases}$$

$$T \equiv 2C\left(\langle x^*\cos\lambda^*\rangle_0 + \langle y^*\sin\lambda^*\rangle_0\right).$$

, h (42)

3. 0 - (42) , .

4.3.1. W^s W^u

$\mathcal{O}(\varepsilon)$ ($|h| \le R(\varepsilon) = \mathcal{O}(\varepsilon)$), μ:

$$h = \mu\varepsilon$$

$$|\mu| \le \frac{R(\varepsilon)}{\varepsilon} = \mathcal{O}(1)$$

(42):

$$\langle H\rangle_0(\boldsymbol{z}) = H_{old}(\boldsymbol{z}) + \hat{H}(\boldsymbol{z}, h, \varepsilon),$$

$$\hat{H}(\boldsymbol{z}, h, \varepsilon) \equiv -h^2\theta_1 H_{old}(\boldsymbol{z}) + 2F\langle x^*\rangle_0 z_x - \langle\cos\lambda^* - 1\rangle_0\Big(U(z_x, z_y)\cos z_\lambda + V(z_x, z_y)\sin z_\lambda\Big) -$$
$$- \cos z_\lambda\Big((2C\hat{x}_0 + e_J D)\langle y^*\sin\lambda^*\rangle_0 + 2Cz_x\left(\langle x^*\cos\lambda^*\rangle + \langle y^*\sin\lambda^*\rangle\right) + \Big\langle\big(U(x^*, y^*) - U_0\big)\cos\lambda^*\Big\rangle_0\Big) -$$
$$- \sin z_\lambda\Big(2Cz_y\left(\langle x^*\cos\lambda^*\rangle + \langle y^*\sin\lambda^*\rangle\right)\Big) + \mathcal{O}(h^4). \quad (43)$$

(43) μ ε:

$$\hat{H}(\boldsymbol{z}, h, \varepsilon) \equiv -\mu^2\varepsilon^2\theta_1 H_{old}(\boldsymbol{z}) + 2F\underbrace{\langle x^*\rangle_0}_{\mu^2\mathcal{O}(\varepsilon^2)} z_x - \underbrace{\langle\cos\lambda^* - 1\rangle_0}_{\mu^2\mathcal{O}(\varepsilon^2)}\Big(U(z_x, z_y)\cos z_\lambda + V(z_x, z_y)\sin z_\lambda\Big) -$$
$$- \cos z_\lambda\Big((2C\hat{x}_0 + e_J D)\underbrace{\langle y^*\sin\lambda^*\rangle_0}_{\mu^2\mathcal{O}(\varepsilon^4)} + 2Cz_x\left(\underbrace{\langle x^*\cos\lambda^*\rangle_0}_{\mu^2\mathcal{O}(\varepsilon^2)} + \underbrace{\langle y^*\sin\lambda^*\rangle_0}_{\mu^2\mathcal{O}(\varepsilon^4)}\right) + \underbrace{\Big\langle\big(U(x^*, y^*) - U_0\big)\cos\lambda^*\Big\rangle_0}_{\mu^2\mathcal{O}(\varepsilon^2)}\Big) -$$
$$- \sin z_\lambda\Big(2Cz_y\left(\underbrace{\langle x^*\cos\lambda^*\rangle_0}_{\mu^2\mathcal{O}(\varepsilon^2)} + \underbrace{\langle y^*\sin\lambda^*\rangle_0}_{\mu^2\mathcal{O}(\varepsilon^4)}\right)\Big) =$$
$$= \varepsilon^2\mu^2\Bigg(- H_{old}(\boldsymbol{z}) + 2Fz_x\underbrace{(-2\varepsilon\xi)\{a\}_{1,1}}_{\mathcal{O}(1)} -$$
$$- \underbrace{\frac{1}{4}\left(\frac{2(\xi^2 - \alpha U_0)}{(2C\hat{x}_0 + e_J D)}\right)^2}_{\mathcal{O}(1)}\Big(U(z_x, z_y)\cos z_\lambda + V(z_x, z_y)\sin z_\lambda\Big) -$$
$$- \cos z_\lambda\left(2Cz_x + (2C\hat{x}_0 + e_J D)\right)\underbrace{(-2\varepsilon\xi)\{a\}_{1,1}}_{\mathcal{O}(1)} - \sin z_\lambda 2Cz_y\underbrace{(-2\varepsilon\xi)\{a\}_{1,1}}_{\mathcal{O}(1)}\Bigg) + \mathcal{O}(\varepsilon^4).$$

$[\cdot]_0$ ε $H_1(\boldsymbol{z})$, μ ε:

$$H_1(\boldsymbol{z}) \equiv \Bigg[- \theta_1 H_{old}(\boldsymbol{z}) + 2Fz_x\underbrace{(-2\varepsilon\xi)\{a\}_{1,1}}_{\mathcal{O}(1)} - \underbrace{\frac{1}{4}\left(\frac{2(\xi^2 - \alpha U_0)}{(2C\hat{x}_0 + e_J D)}\right)^2}_{\mathcal{O}(1)}\Big(U(z_x, z_y)\cos z_\lambda + V(z_x, z_y)\sin z_\lambda\Big) -$$
$$- \cos z_\lambda\left(2Cz_x + (2C\hat{x}^0 + e_J D)\right)\underbrace{(-2\varepsilon\xi)\{a\}_{1,1}}_{\mathcal{O}(1)} - \sin z_\lambda 2Cz_y\underbrace{(-2\varepsilon\xi)\{a\}_{1,1}}_{\mathcal{O}(1)}\Bigg]_0.$$

:

$$\langle H \rangle_0 = H_{old}(\boldsymbol{z}) + \varepsilon^2 \mu^2 H_1(\boldsymbol{z}) + \mathcal{O}(\varepsilon^4). \tag{44}$$

$$\hat{H}(\boldsymbol{z}) = \varepsilon^2 \mu^2 H_1(\boldsymbol{z}) + \mathcal{O}(\varepsilon^4).$$

(44) (22):

$$\begin{cases} z_\Lambda^{s,u}(\tau,\mu) = \sum_{k=0}^{\infty} \varepsilon^k z_{\Lambda,k}^{s,u}(\tau,\mu), \\ z_\lambda^{s,u}(\tau,\mu) = \sum_{k=0}^{\infty} \varepsilon^k z_{\lambda,k}^{s,u}(\tau,\mu), \\ z_x^{s,u}(\tau,\mu) = \sum_{k=0}^{\infty} \varepsilon^k z_{x,k}^{s,u}(\tau,\mu), \\ z_y^{s,u}(\tau,\mu) = \sum_{k=0}^{\infty} \varepsilon^k z_{y,k}^{s,u}(\tau,\mu), \end{cases} \tag{45}$$

:

- $W^s(0)$:

$$\begin{cases} z_\Lambda^s(+\infty) = 0, \\ z_\lambda^s(+\infty) = 0, \\ z_x^s(+\infty) = 0, \\ z_y^s(+\infty) = 0, \end{cases}$$

- $W^u(0)$:

$$\begin{cases} z_\Lambda^u(-\infty) = 0, \\ z_\lambda^u(-\infty) = 0, \\ z_x^u(-\infty) = 0, \\ z_y^u(-\infty) = 0. \end{cases}$$

(45) (44)

$$\begin{cases} \dot{z}_\Lambda = -U \sin z_\lambda + V \cos z_\lambda - \varepsilon^2 \mu^2 \dfrac{\partial H_1}{\partial z_\lambda} + \mathcal{O}(\varepsilon^4), \\ \dot{z}_\lambda = \alpha z_\Lambda + \varepsilon^2 \mu^2 \dfrac{\partial H_1}{\partial z_\Lambda} + \mathcal{O}(\varepsilon^4), \\ \dot{z}_x = -\varepsilon \left(\left(2F z_y - \dfrac{\partial U}{\partial z_y} \cos z_\lambda - \dfrac{\partial V}{\partial z_y} \sin z_\lambda\right) + \varepsilon^2 \mu^2 \dfrac{\partial H_1}{\partial z_y} + \mathcal{O}(\varepsilon^4) \right), \\ \dot{z}_y = \varepsilon \left(\left(2F(z_x + \hat{x}_0) + e_J G - \dfrac{\partial U}{\partial z_x} \cos z_\lambda - \dfrac{\partial V}{\partial z_x} \sin z_\lambda\right) + \varepsilon^2 \mu^2 \dfrac{\partial H_1}{\partial x} + \mathcal{O}(\varepsilon^4) \right), \end{cases} \tag{46}$$

ε.

. :

$$\begin{cases} z_{\Lambda,0}(\tau) = 0, \\ z_{\lambda,0}(\tau) = \lambda_-(x_{sep}(\tau), y_{sep}(\tau)), \\ z_{x,0}(\tau) = x_{sep}(\tau) - \hat{x}_0, \\ z_{y,0}(\tau) = y_{sep}(\tau). \end{cases}$$

$(z_{\Lambda,1}, z_{\lambda,1}, z_{x,1}, z_{y,1})$ $(z_{\Lambda,1}, z_{\lambda,1})$:

$$\begin{cases} z_{\Lambda,1}(\tau) = z'_{\lambda,0}, \\ z_{\lambda,1}(\tau) = \frac{1}{\beta}\Big(z_{x,1}\big((2Cz_{x,0} + 2C\hat{x}_0 + e_J D)\sin z_{\lambda,0} - 2Cz_{y,0}\cos z_{\lambda,0}\big) + \\ + z_{y,1}\big(-2Cz_{y,0}\sin z_{\lambda,0} - (2Cz_{x,0} + 2C\hat{x}_0 + e_J D)\cos z_{\lambda,0}\big)\Big), \end{cases}$$

$(z_{x,1}, z_{y,1})$:

$$\frac{d}{d\tau}\begin{pmatrix} z_{x,1} \\ z_{y,1} \end{pmatrix} = \mathcal{A}(\tau)\begin{pmatrix} z_{x,1} \\ z_{y,1} \end{pmatrix}$$

$\mathcal{A}(\tau)$ - (24). - $\boldsymbol{u}_1(\tau), \tilde{\boldsymbol{u}}_{\boldsymbol{1}}(\tau)$.
10 W^s W^u , . :

$$(z_{\Lambda,1}(\tau), z_{\lambda,1}(\tau), z_{x,1}(\tau), z_{y,1}(\tau)) \equiv (0,0,0,0)$$

, μ, μ .
$(z_{x,2}, z_{y,2})$, μ:

$$\frac{d}{d\tau}\begin{pmatrix} z_{x,2} \\ z_{y,2} \end{pmatrix} = \mathcal{A}(\tau)\begin{pmatrix} z_{x,2} \\ z_{y,2} \end{pmatrix} + \begin{pmatrix} \tilde{G}_2^x(\tau) \\ \tilde{G}_2^y(\tau) \end{pmatrix} + \mu^2 \begin{pmatrix} \mathfrak{G}_2^x(\tau) \\ \mathfrak{G}_2^y(\tau) \end{pmatrix}, \tag{47}$$

:

$$\mathfrak{G}_2^x = \frac{\partial H_1}{\partial z_y}\Big|_{z_{\Lambda,0}, z_{\lambda,0}, z_{x,0}, z_{y,0}} - \\ - \beta^{-1}\left(\frac{\partial H_1}{\partial z_\lambda}\Big|_{z_{\Lambda,0}, z_{\lambda,0}, z_{x,0}, z_{y,0}}\right)\big(-2Cz_{y,0}\sin z_{\lambda,0} - (2Cz_{x,0} + 2C\hat{x}_0 + e_J D)\cos z_{\lambda,0}\big),$$

$$\mathfrak{G}_2^y = \frac{\partial H_1}{\partial z_x}\Big|_{z_{\Lambda,0}, z_{\lambda,0}, z_{x,0}, z_{y,0}} - \\ - \beta^{-1}\left(\frac{\partial H_1}{\partial z_\lambda}\Big|_{z_{\Lambda,0}, z_{\lambda,0}, z_{x,0}, z_{y,0}}\right)\big(-(2Cz_{x,0} + 2C\hat{x}_0 + e_J D)\sin z_{\lambda,0} + 2Cz_{y,0}\cos z_{\lambda,0}\big).$$

$$\begin{cases} z_{\Lambda,2}(\tau) = \dfrac{1}{\alpha} z'_{\lambda,1}(\tau), \\ z_{\lambda,2}(\tau) = \dfrac{1}{\beta}\Big(\alpha^{-1} z''_{\lambda,0} + \dfrac{\partial H_1}{\partial z_\lambda}\Big|_{z_{\Lambda,0}, z_{\lambda,0}, z_{x,0}, z_{y,0}} + \\ + z_{x,2}\big((2Cz_{x,0} + 2C\hat{x}_0 + e_J D)\sin z_{\lambda,0} - 2Cz_{y,0}\cos z_{\lambda,0}\big) + \\ + z_{y,2}\big(-2Cz_{y,0}\sin z_{\lambda,0} - (2Cz_{x,0} + 2C\hat{x}_0 + e_J D)\cos z_{\lambda,0}\big)\Big). \end{cases}$$

, $\mathfrak{G}_2^x(\tau)$ – , $\mathfrak{G}_2^y(\tau)$ – .
, $\tilde{G}_2^x, \tilde{G}_2^y$ $\boldsymbol{z}_0, \boldsymbol{z}_1$:

$$\tilde{G}_2^x = -\beta^{-1}\left(\alpha^{-1} z''_{\lambda,0}\right)\big(-2Cz_{y,0}\sin z_{\lambda,0} - (2Cz_{x,0} + 2C\hat{x}_0 + e_J D)\cos z_{\lambda,0}\big),$$

$$\tilde{G}_2^y = -\beta^{-1}\left(\alpha^{-1} z''_{\lambda,0}\right)\big(-(2Cz_{x,0} + 2C\hat{x}_0 + e_J D)\sin z_{\lambda,0} + 2Cz_{y,0}\cos z_{\lambda,0}\big).$$

, (47) (24) , (47) :

$$\begin{cases} z_{x,2}(\tau,\mu) = z^0_{x,2}(\tau) + \hat{z}_{x,2}(\tau,\mu), \\ z_{y,2}(\tau,\mu) = z^0_{y,2}(\tau) + \hat{z}_{y,2}(\tau,\mu), \end{cases}$$

$(z^0_{x,2}, z^0_{y,2})$ - (24), $(\hat{z}_{x,2}, \hat{z}_{y,2})$ - :

$$\frac{d}{d\tau}\begin{pmatrix} \hat{z}_{x,2} \\ \hat{z}_{y,2} \end{pmatrix} = \mathcal{A}(\tau)\begin{pmatrix} \hat{z}_{x,2} \\ \hat{z}_{y,2} \end{pmatrix} + \mu^2 \begin{pmatrix} \mathfrak{G}_2^x(\tau) \\ \mathfrak{G}_2^y(\tau) \end{pmatrix},$$

$\boldsymbol{u}_2 = (x_2, y_2)$, $\boldsymbol{u}_2^0 = (x_2^0, y_2^0)$, $\hat{\boldsymbol{u}}_{\boldsymbol{2}} = (\hat{x}_2, \hat{y}_2)$. $\hat{\boldsymbol{u}}_{\boldsymbol{2}}^{\boldsymbol{s},\boldsymbol{u}}$ (s u):

$$\begin{cases} \hat{\boldsymbol{u}}_{\boldsymbol{2}}^{\boldsymbol{u}} = \mu^2 \boldsymbol{u}_1(\tau) \displaystyle\int_{\tau^u}^{\tau} (\tilde{y}_1(\tau)\mathfrak{G}_2^x(\tau) - \tilde{x}_1(\tau)\mathfrak{G}_2^y(\tau))\, d\tau + \mu^2 \tilde{\boldsymbol{u}}_{\boldsymbol{1}}(\tau) \displaystyle\int_{-\infty}^{\tau} (x_1(\tau)\mathfrak{G}_2^y(\tau) - y_1(\tau)\mathfrak{G}_2^x(\tau))\, d\tau, \\ \\ \hat{\boldsymbol{u}}_{\boldsymbol{2}}^{\boldsymbol{s}} = \mu^2 \boldsymbol{u}_1(\tau) \displaystyle\int_{\tau^s}^{\tau} (\tilde{y}_1(\tau)\mathfrak{G}_2^x(\tau) - \tilde{x}_1(\tau)\mathfrak{G}_2^y(\tau))\, d\tau - \mu^2 \tilde{\boldsymbol{u}}_{\boldsymbol{1}}(\tau) \displaystyle\int_{\tau}^{+\infty} (x_1(\tau)\mathfrak{G}_2^y(\tau) - y_1(\tau)\mathfrak{G}_2^x(\tau))\, d\tau. \end{cases}$$

, $W^u(\Gamma_{h_1})$ $W^s(\Gamma_{h_2})$,

$$\boldsymbol{u}_2^u(\tau_1, \mu_1) - \boldsymbol{u}_2^s(\tau_2, \mu_2) = \left(\boldsymbol{u}_2^{0,u}(\tau_1) - \boldsymbol{u}_2^{0,s}(\tau_2)\right) + \left(\hat{\boldsymbol{u}}_{\boldsymbol{2}}^{\boldsymbol{u}}(\tau_1, \mu_1) - \hat{\boldsymbol{u}}_{\boldsymbol{2}}^{\boldsymbol{s}}(\tau_2, \mu_2)\right). \tag{48}$$

$\hat{\boldsymbol{u}}_{\boldsymbol{2}}^{\boldsymbol{u}}, \hat{\boldsymbol{u}}_{\boldsymbol{2}}^{\boldsymbol{s}}$ (48) :

$$B(\tau_1, \tau_2) \cdot \begin{pmatrix} \mu_1^2 \\ \mu_2^2 \end{pmatrix} = \begin{pmatrix} f_1(\tau_1, \tau_2) \\ f_2(\tau_1, \tau_2) \end{pmatrix}, \tag{49}$$

$$B_{1,1} = x_1(\tau_1) \int_{\tau^u}^{\tau_1} (\tilde{y}_1 \mathfrak{G}_2^x - \tilde{x}_1 \mathfrak{G}_2^y)\, d\tau + \tilde{x}_1(\tau_1) \int_{-\infty}^{\tau_1} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau,$$

$$B_{1,2} = -x_1(\tau_2) \int_{\tau^s}^{\tau_2} (\tilde{y}_1 \mathfrak{G}_2^x - \tilde{x}_1 \mathfrak{G}_2^y)\, d\tau + \tilde{x}_1(\tau_2) \int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau,$$

$$B_{2,1} = y_1(\tau_1) \int_{\tau^u}^{\tau_1} (\tilde{y}_1 \mathfrak{G}_2^x - \tilde{x}_1 \mathfrak{G}_2^y)\, d\tau + \tilde{y}_1(\tau_1) \int_{-\infty}^{\tau_1} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau,$$

$$B_{2,2} = -y_1(\tau_2) \int_{\tau^s}^{\tau_2} (\tilde{y}_1 \mathfrak{G}_2^x - \tilde{x}_1 \mathfrak{G}_2^y)\, d\tau + \tilde{y}_1(\tau_2) \int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau,$$

$$\begin{aligned} f_1 = -x_2^{0,u}(\tau_1) + x_2^{0,s}(\tau_1) = \\ = -x_1(\tau_1) \int_{\tau_2^u}^{\tau_1} \left(\tilde{y}_1 \tilde{G}_2^x - \tilde{x}_1 \tilde{G}_2^y\right) d\tau - \tilde{x}_1(\tau_1) \int_{-\infty}^{\tau_1} \left(x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x\right) d\tau + \\ + x_1(\tau_2) \int_{\tau_2^s}^{\tau_2} \left(\tilde{y}_1 \tilde{G}_2^x - \tilde{x}_1 \tilde{G}_2^y\right) d\tau - \tilde{x}_1(\tau_2) \int_{\tau_2}^{+\infty} \left(x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x\right) d\tau, \end{aligned}$$

$$\begin{aligned} f_2 = -y_2^{0,u}(\tau_1) + y_2^{0,s}(\tau_1) = \\ = -y_1(\tau_1) \int_{\tau_2^u}^{\tau_1} \left(\tilde{y}_1 \tilde{G}_2^x - \tilde{x}_1 \tilde{G}_2^y\right) d\tau - \tilde{y}_1(\tau_1) \int_{-\infty}^{\tau_1} \left(x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x\right) d\tau + \\ + y_1(\tau_2) \int_{\tau_2^s}^{\tau_2} \left(\tilde{y}_1 \tilde{G}_2^x - \tilde{x}_1 \tilde{G}_2^y\right) d\tau - \tilde{y}_1(\tau_2) \int_{\tau_2}^{+\infty} \left(x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x\right) d\tau. \end{aligned}$$

, (49) μ_1^2, μ_2^2, $\{\tau_1, \tau_2, \tau^s, \tau^u, \tau_2^s, \tau_2^u\}$.
(49) $\det B(\tau_1, \tau_2) \neq 0$.
$\tau^u = \tau_1$, $\tau^s = \tau_2$, :

$$\begin{aligned} \det B(\tau_1, \tau_2) = \int_{-\infty}^{\tau_1} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau \int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau\, (\tilde{x}_1(\tau_1)\tilde{y}_1(\tau_2) - \tilde{x}_1(\tau_2)\tilde{y}_1(\tau_1)) \equiv \\ \equiv F(\tau_1, \tau_2) \int_{-\infty}^{\tau_1} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau \int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau, \end{aligned}$$

$$B_{1,1} = \tilde{x}_1(\tau_1) \int_{-\infty}^{\tau_1} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau,$$

$$B_{1,2} = \tilde{x}_1(\tau_2) \int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau,$$

$$B_{2,1} = \tilde{y}_1(\tau_1) \int_{-\infty}^{\tau_1} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau,$$

$$B_{2,2} = \tilde{y}_1(\tau_2) \int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau.$$

, $\mu_1, \mu_2 > 0,$ (49) $\mu_1^2 > 0, \mu_2^2 > 0.$:

$$\frac{B_{1,1} f_2 - B_{2,1} f_1}{\det B} > 0,$$
$$\frac{B_{2,2} f_1 - B_{1,2} f_2}{\det B} > 0,$$

$$B_{1,1} f_2 - B_{2,1} f_1 = \int_{-\infty}^{\tau_1} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau \Bigg(\int_{\tau_2^u}^{\tau_1} \left(\tilde{y}_1 \tilde{G}_2^x - \tilde{x}_1 \tilde{G}_2^y \right) d\tau \Big(\underbrace{x_1(\tau_1)\tilde{y}_1(\tau_1) - \tilde{x}_1(\tau_1) y_1(\tau_1)}_{W(\boldsymbol{u}_1, \tilde{\boldsymbol{u}}_1) = 1} \Big) + \int_{\tau_2^s}^{\tau_2} \left(\tilde{y}_1 \tilde{G}_2^x - \tilde{x}_1 \tilde{G}_2^y \right) d\tau \Big(\tilde{x}_1(\tau_1) y_1(\tau_2) - \tilde{y}_1(\tau_1) x_1(\tau_2) \Big) + \int_{\tau_2}^{+\infty} \left(x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x \right) d\tau \Big(\underbrace{\tilde{y}_1(\tau_1)\tilde{x}_1(\tau_2) - \tilde{x}_1(\tau_1)\tilde{y}_1(\tau_2)}_{-F(\tau_1, \tau_2)} \Big) \Bigg),$$

$$B_{2,2} f_1 - B_{1,2} f_2 = \int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau \Bigg(\int_{\tau_2^u}^{\tau_1} \left(\tilde{y}_1 \tilde{G}_2^x - \tilde{x}_1 \tilde{G}_2^y \right) d\tau \Big(\tilde{x}_1(\tau_2) y_1(\tau_1) - \tilde{y}_1(\tau_2) x_1(\tau_1) \Big) + \int_{\tau_2^s}^{\tau_2} \left(\tilde{y}_1 \tilde{G}_2^x - \tilde{x}_1 \tilde{G}_2^y \right) d\tau \Big(\underbrace{x_1(\tau_2)\tilde{y}_1(\tau_2) - \tilde{x}_1(\tau_2) y_1(\tau_2)}_{W(\boldsymbol{u}_1, \tilde{\boldsymbol{u}}_1) = 1} \Big) + \int_{-\infty}^{\tau_1} \left(x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x \right) d\tau \Big(\underbrace{\tilde{y}_1(\tau_1)\tilde{x}_1(\tau_2) - \tilde{x}_1(\tau_1)\tilde{y}_1(\tau_2)}_{-F(\tau_1, \tau_2)} \Big) \Bigg).$$

, :

$$\frac{B_{1,1} f_2 - B_{2,1} f_1}{\det B} = -\frac{\int_{\tau_2}^{+\infty} \left(x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x \right) d\tau}{\int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau} + \frac{\int_{\tau_2^u}^{\tau_1} \left(\tilde{x}_1 \tilde{G}_2^y - \tilde{y}_1 \tilde{G}_2^x \right) d\tau}{F(\tau_1, \tau_2) \int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau} + \frac{\int_{\tau_2^s}^{\tau_2} \left(\tilde{x}_1 \tilde{G}_2^y - \tilde{y}_1 \tilde{G}_2^x \right) d\tau \left(\tilde{x}_1(\tau_1) y_1(\tau_2) - \tilde{y}_1(\tau_1) x_1(\tau_2) \right)}{F(\tau_1, \tau_2) \int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x)\, d\tau} > 0,$$

$$\frac{B_{2,2}f_1 - B_{1,2}f_2}{\det B} = -\frac{\int_{-\infty}^{\tau_1}\left(x_1\tilde{G}_2^y - y_1\tilde{G}_2^x\right)d\tau}{\int_{-\infty}^{\tau_1}\left(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x\right)d\tau} + \frac{\int_{\tau_2^s}^{\tau_2}\left(\tilde{x}_1\tilde{G}_2^y - \tilde{y}_1\tilde{G}_2^x\right)d\tau}{F(\tau_1,\tau_2)\int_{-\infty}^{\tau_1}\left(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x\right)d\tau} +$$

$$+ \frac{\int_{\tau_2^u}^{\tau_1}\left(\tilde{x}_1\tilde{G}_2^y - \tilde{y}_1\tilde{G}_2^x\right)d\tau\,\left(\tilde{x}_1(\tau_2)y_1(\tau_1) - \tilde{y}_1(\tau_2)x_1(\tau_1)\right)}{F(\tau_1,\tau_2)\int_{-\infty}^{\tau_1}\left(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x\right)d\tau} > 0,$$

$$F(\tau_1,\tau_2) \equiv \tilde{y}_1(\tau_1)\tilde{x}_1(\tau_2) - \tilde{x}_1(\tau_1)\tilde{y}_1(\tau_2).$$

5.

6.

6.1. 1

1) $g^\rho(\lambda,\Lambda,x,y)$ (22). :

$$g^\rho(\lambda,\Lambda,x,y) = g^\rho(\lambda_0,\Lambda_0,x_0,y_0) + \sum_{j=1}^{+\infty}\frac{\mathcal{T}^j g^\rho}{j!},$$

$\mathcal{T}$ - :

$$\mathcal{T} = \left(\sum_{k=1}^{+\infty}\lambda_k\varepsilon^k\right)\frac{\partial}{\partial\lambda}\bigg|_{\lambda_0,\Lambda_0,x_0,y_0} + \left(\sum_{k=1}^{+\infty}\Lambda_k\varepsilon^k\right)\frac{\partial}{\partial\Lambda}\bigg|_{\lambda_0,\Lambda_0,x_0,y_0} +$$

$$+ \left(\sum_{k=1}^{+\infty}x_k\varepsilon^k\right)\frac{\partial}{\partial x}\bigg|_{\lambda_0,\Lambda_0,x_0,y_0} + \left(\sum_{k=1}^{+\infty}y_k\varepsilon^k\right)\frac{\partial}{\partial y}\bigg|_{\lambda_0,\Lambda_0,x_0,y_0}.$$

$$\mathfrak{D}_k = \left(\begin{pmatrix}\lambda_k\\ \Lambda_k\\ x_k\\ y_k\end{pmatrix}, \begin{pmatrix}\frac{\partial}{\partial\lambda}\big|_{\lambda_0,\Lambda_0,x_0,y_0}\\ \frac{\partial}{\partial\Lambda}\big|_{\lambda_0,\Lambda_0,x_0,y_0}\\ \frac{\partial}{\partial x}\big|_{\lambda_0,\Lambda_0,x_0,y_0}\\ \frac{\partial}{\partial y}\big|_{\lambda_0,\Lambda_0,x_0,y_0}\end{pmatrix}\right),$$

:

$$\mathcal{T} = \sum_{k=1}^{+\infty}\varepsilon^k\mathfrak{D}_k.$$

$\mathcal{T}^j$, , :

$$\mathcal{T}^j = \left(\sum_{k=1}^{+\infty}\varepsilon^k\mathfrak{D}_k\right)^j = \sum_{k=j}^{+\infty}\varepsilon^k\mathcal{L}_k^{(j)},$$

:

$$\mathcal{L}_k^{(j)} = \sum_{k_1+\ldots+k_j=k}\mathfrak{D}_{k_1}\cdot\ldots\cdot\mathfrak{D}_{k_j}.$$

ε :

$$g^\rho(\lambda,\Lambda,x,y) = g^\rho(\lambda_0,\Lambda_0,x_0,y_0) + \sum_{k=1}^{+\infty}\varepsilon^k\left(\sum_{j=1}^{k}\frac{\mathcal{L}_k^{(j)}g^\rho}{j!}\right).$$

2 , $\mathcal{L}_k^{(1)} = \mathfrak{D}_k$:

$$g^\rho(\lambda,\Lambda,x,y) = g^\rho(\lambda_0,\Lambda_0,x_0,y_0) + \varepsilon\mathfrak{D}_1 g^\rho + \sum_{k=2}^{+\infty}\varepsilon^k\left(\mathcal{L}_k^{(1)}g^\rho + \sum_{j=2}^{k}\frac{\mathcal{L}_k^{(j)}g^\rho}{j!}\right) =$$

$$= g^\rho(\lambda_0,\Lambda_0,x_0,y_0) + \varepsilon\mathfrak{D}_1 g^\rho + \sum_{k=2}^{+\infty}\varepsilon^k\left(\mathfrak{D}_k g^\rho + \underbrace{\sum_{j=2}^{k}\frac{\mathcal{L}_k^{(j)}g^\rho}{j!}}_{G_k^\rho}\right)$$

- $\mathfrak{D}_k g^\rho$ - $(\lambda_k,\Lambda_k,x_k,y_k)$;
- G_k^ρ - $(\lambda_j,\Lambda_j,x_j,y_j)$, $j \le k-1$ (24).

G_k^ρ $k \ge 2$, $k = 1$ $G_1^\rho = 0$. :

$$G_k^\rho = \sum_{j=2}^{k}\frac{\mathcal{L}_k^{(j)}g^\rho}{j!}, \quad k \ge 2, \tag{50}$$

$$G_1^\rho \equiv 0,$$

$$g^\rho(\lambda,\Lambda,x,y) = g^\rho(\lambda_0,\Lambda_0,x_0,y_0) + \sum_{k=1}^{\infty}\varepsilon^k\mathfrak{D}_k g^\rho + \sum_{k=2}^{+\infty}\varepsilon^k G_k^\rho.$$

2) $g^\rho, \rho \in \{\lambda,\Lambda,x,y\}$, (50) $j = 2$, G_{k+1}^ρ 2 $(\le k)$ $\mathfrak{D}_i\mathfrak{D}_j g^\rho, i+j \le k$. $\le k$ $\mathcal{F}_s$, , , G_{k+1}^ρ $\mathcal{F}_{2s}$.

3) $g^\lambda = \alpha\Lambda$, 0. $G_k^\lambda = 0$ $\forall k \ge 1$. ■

6.2.

1) 4 - :

$$\begin{cases}\dot\Lambda(t) = g^\Lambda(\Lambda,\lambda,x,y),\\ \dot\lambda(t) = g^\lambda(\Lambda,\lambda,x,y),\\ \dot x(t) = \varepsilon g^x(\Lambda,\lambda,x,y),\\ \dot y(t) = \varepsilon g^y(\Lambda,\lambda,x,y),\end{cases}$$

$\tau = \varepsilon t$:

$$\begin{cases}\varepsilon\Lambda'(\tau) = g^\Lambda(\Lambda,\lambda,x,y),\\ \varepsilon\lambda'(\tau) = g^\lambda(\Lambda,\lambda,x,y),\\ x'(\tau) = g^x(\Lambda,\lambda,x,y),\\ y'(\tau) = g^y(\Lambda,\lambda,x,y),\end{cases}$$

$\boldsymbol{X} = (\Lambda, \lambda, x, y)$: $\boldsymbol{X} = \sum_{k=0}^{\infty} \varepsilon^k \boldsymbol{X}_k$. ε :

$$\begin{cases} 0 = g^{\Lambda}(\boldsymbol{X}_0), \\ 0 = g^{\lambda}(\boldsymbol{X}_0), \\ x_0'(\tau) = g^{x}(\boldsymbol{X}_0), \\ y_0'(\tau) = g^{y}(\boldsymbol{X}_0), \end{cases} \tag{51}$$

$$\begin{cases} \Lambda_0'(\tau) = (\nabla g^{\Lambda}(\boldsymbol{X}_0), \boldsymbol{X}_1), \\ \lambda_0'(\tau) = (\nabla g^{\lambda}(\boldsymbol{X}_0), \boldsymbol{X}_1), \\ x_1'(\tau) = (\nabla g^{x}(\boldsymbol{X}_0), \boldsymbol{X}_1), \\ y_1'(\tau) = (\nabla g^{y}(\boldsymbol{X}_0), \boldsymbol{X}_1), \end{cases} \tag{52}$$

, (52) (Λ_1, λ_1), :

$$\begin{pmatrix} (\nabla g^{\Lambda})_{\Lambda} & (\nabla g^{\Lambda})_{\lambda} \\ (\nabla g^{\lambda})_{\Lambda} & (\nabla g^{\lambda})_{\lambda} \end{pmatrix} \begin{pmatrix} \Lambda_1 \\ \lambda_1 \end{pmatrix} = A \begin{pmatrix} \Lambda_1 \\ \lambda_1 \end{pmatrix} = \begin{pmatrix} \Lambda_0' - (\nabla g^{\Lambda})_x x_1 - (\nabla g^{\Lambda})_y y_1 \\ \lambda_0' - (\nabla g^{\lambda})_x x_1 - (\nabla g^{\lambda})_y y_1 \end{pmatrix},$$

$$A = \begin{pmatrix} (\nabla g^{\Lambda})_{\Lambda} & (\nabla g^{\Lambda})_{\lambda} \\ (\nabla g^{\lambda})_{\Lambda} & (\nabla g^{\lambda})_{\lambda} \end{pmatrix}.$$

:

$$\begin{pmatrix} \Lambda_1 \\ \lambda_1 \end{pmatrix} = M \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \boldsymbol{C},$$

$$M = \frac{1}{\det A} \begin{pmatrix} -(\nabla g^{\lambda})_{\lambda}(\nabla g^{\Lambda})_x + (\nabla g^{\Lambda})_{\lambda}(\nabla g^{\lambda})_x & -(\nabla g^{\lambda})_{\lambda}(\nabla g^{\Lambda})_y + (\nabla g^{\Lambda})_{\lambda}(\nabla g^{\lambda})_y \\ (\nabla g^{\lambda})_{\Lambda}(\nabla g^{\Lambda})_x - (\nabla g^{\Lambda})_{\Lambda}(\nabla g^{\lambda})_x & (\nabla g^{\lambda})_{\Lambda}(\nabla g^{\Lambda})_y - (\nabla g^{\Lambda})_{\Lambda}(\nabla g^{\lambda})_y \end{pmatrix},$$

$$\boldsymbol{C} = \frac{1}{\det A} \begin{pmatrix} (\nabla g^{\lambda})_{\lambda}\Lambda_0' - (\nabla g^{\Lambda})_{\lambda}\lambda_0' \\ -(\nabla g^{\lambda})_{\Lambda}\Lambda_0' + (\nabla g^{\Lambda})_{\Lambda}\lambda_0' \end{pmatrix}.$$

(52) :

$$\begin{cases} x_1'(\tau) = \Big((\nabla g^{x})_x + \big((\nabla g^{x})_{\Lambda} M_{11} + (\nabla g^{x})_{\lambda} M_{21}\big)\Big) x_1 + \\ \qquad + \Big((\nabla g^{x})_y + \big((\nabla g^{x})_{\Lambda} M_{12} + (\nabla g^{x})_{\lambda} M_{22}\big)\Big) y_1 + \\ \qquad + \left(\begin{pmatrix} (\nabla g^{x})_{\Lambda} \\ (\nabla g^{x})_{\lambda} \end{pmatrix}, \boldsymbol{C} \right), \\ \\ y_1'(\tau) = \Big((\nabla g^{y})_x + \big((\nabla g^{y})_{\Lambda} M_{11} + (\nabla g^{y})_{\lambda} M_{21}\big)\Big) x_1 + \\ \qquad + \Big((\nabla g^{x})_y + \big((\nabla g^{y})_{\Lambda} M_{12} + (\nabla g^{y})_{\lambda} M_{22}\big)\Big) y_1 + \\ \qquad + \left(\begin{pmatrix} (\nabla g^{y})_{\Lambda} \\ (\nabla g^{y})_{\lambda} \end{pmatrix}, \boldsymbol{C} \right). \end{cases} \tag{53}$$

$(\cdot,\cdot)$ – $\mathbb{R}^n$.

, , , (20) . , :

$$\Lambda_0 = 0 \quad \Rightarrow \quad \boldsymbol{C} = \begin{pmatrix} \frac{\lambda_0'}{\alpha} \\ 0 \end{pmatrix}$$

, g^x, g^y Λ, , (53) :

$$\begin{pmatrix} (\nabla g^x)_\Lambda \\ (\nabla g^x)_\lambda \end{pmatrix} = \begin{pmatrix} 0 \\ * \end{pmatrix},$$
$$\begin{pmatrix} (\nabla g^y)_\Lambda \\ (\nabla g^y)_\lambda \end{pmatrix} = \begin{pmatrix} 0 \\ * \end{pmatrix}.$$

(53) :

$$\left(\begin{pmatrix} (\nabla g^x)_\Lambda \\ (\nabla g^x)_\lambda \end{pmatrix}, \boldsymbol{C} \right) = \left(\begin{pmatrix} (\nabla g^y)_\Lambda \\ (\nabla g^y)_\lambda \end{pmatrix}, \boldsymbol{C} \right) = 0.$$

, (20) (x_1, y_1) .

2) (51). x_0.

x_0 , , Λ_0, λ_0 x_0, y_0:

$$\begin{aligned} x_0'' = (\nabla g^x)_x x_0' + (\nabla g^x)_y y_0' + (\nabla g^x)_\Lambda \Lambda_0' + (\nabla g^x)_\lambda \lambda_0' = \\ = \left((\nabla g^x)_x + (\nabla g^x)_\Lambda \frac{\partial \Lambda_0}{\partial x} + (\nabla g^x)_\lambda \frac{\partial \lambda_0}{\partial x} \right) x_0' + \\ + \left((\nabla g^x)_y + (\nabla g^x)_\Lambda \frac{\partial \Lambda_0}{\partial y} + (\nabla g^x)_\lambda \frac{\partial \lambda_0}{\partial y} \right) y_0'. \end{aligned} \tag{54}$$

, $\Lambda_0(x_0, y_0), \lambda_0(x_0, y_0)$:

$$\begin{cases} 0 = g^\Lambda(\boldsymbol{X}_0), \\ 0 = g^\lambda(\boldsymbol{X}_0). \end{cases} \tag{55}$$

x:

$$\begin{cases} (\nabla g^\Lambda)_x + (\nabla g^\Lambda)_y \underbrace{y_x}_{0} + (\nabla g^\Lambda)_\Lambda \cdot \Lambda_x + (\nabla g^\Lambda)_\lambda \cdot \lambda_x = 0, \\ (\nabla g^\lambda)_x + (\nabla g^\lambda)_y \underbrace{y_x}_{0} + (\nabla g^\lambda)_\Lambda \cdot \Lambda_x + (\nabla g^\lambda)_\lambda \cdot \lambda_x = 0. \end{cases}$$

:

$$\begin{pmatrix} (\nabla g^\Lambda)_\Lambda & (\nabla g^\Lambda)_\lambda \\ (\nabla g^\lambda)_\Lambda & (\nabla g^\lambda)_\lambda \end{pmatrix} \begin{pmatrix} \Lambda_x \\ \lambda_x \end{pmatrix} = - \begin{pmatrix} (\nabla g^\Lambda)_x \\ (\nabla g^\lambda)_x \end{pmatrix}.$$

(Λ_x, λ_x) :

$$\Lambda_x = \frac{(\nabla g^\Lambda)_\lambda (\nabla g^\lambda)_x - (\nabla g^\Lambda)_x (\nabla g^\lambda)_\lambda}{(\nabla g^\Lambda)_\Lambda (\nabla g^\lambda)_\lambda - (\nabla g^\Lambda)_\lambda (\nabla g^\lambda)_\Lambda},$$
$$\lambda_x = \frac{(\nabla g^\Lambda)_x (\nabla g^\lambda)_\Lambda - (\nabla g^\Lambda)_\Lambda (\nabla g^\lambda)_x}{(\nabla g^\Lambda)_\Lambda (\nabla g^\lambda)_\lambda - (\nabla g^\Lambda)_\lambda (\nabla g^\lambda)_\Lambda}.$$

, M:

$$\frac{\partial \Lambda_0}{\partial x} = M_{11},$$
$$\frac{\partial \lambda_0}{\partial x} = M_{21},$$

:

$$\frac{\partial \Lambda_0}{\partial y} = M_{12},$$
$$\frac{\partial \lambda_0}{\partial y} = M_{22}.$$

(54) :

$$\begin{aligned}(x_0')' = (\nabla g^x)_x x_0' + (\nabla g^x)_y y_0' + (\nabla g^x)_\Lambda \Lambda_0' + (\nabla g^x)_\lambda \lambda_0' = \\ = ((\nabla g^x)_x + (\nabla g^x)_\Lambda M_{11} + (\nabla g^x)_\lambda M_{21})\, x_0' + \\ + ((\nabla g^x)_y + (\nabla g^x)_\Lambda M_{12} + (\nabla g^x)_\lambda M_{22})\, y_0'. \quad (56)\end{aligned}$$

, (20) (53) , (56) (53) .

, $\tau \to \pm\infty$:

$$\begin{cases} x_0(\tau) = \dfrac{\text{const}}{\cosh(f_\pm \cdot \tau)} + \mathcal{O}(e^{-2f_\pm|\tau|}), \quad \tau \to \pm\infty, \\ y_0(\tau) = \dfrac{\text{const} \cdot \sinh(f_\pm \cdot \tau)}{\cosh^2(f_\pm \cdot \tau)} + \mathcal{O}(e^{-2f_\pm|\tau|}), \quad \tau \to \pm\infty, \end{cases}$$

$f_\pm = \lim_{\tau\to\pm\infty} \frac{df^\pm(\tau)}{d\tau} = \frac{J^+\chi^+}{\sigma^+} \lim_{\tau\to\pm\infty} \frac{P^+(f^+(\tau))}{S^+(f^+(\tau))}$. $P^\pm$ $S^\pm$ $\cosh f^\pm$, :

$$f_+ = f_- = \frac{J^+\chi^+}{\sigma^+} \cdot \frac{1}{U_0} > 0$$

:

$$x_0, y_0 \in \mathcal{F}_{s_0}, \quad s_0 = f_+ = f_- > 0$$

:

$$\begin{cases} x_1(\pm\infty) = 0, \\ y_1(\pm\infty) = 0. \end{cases}$$
$$x_1, y_1 \in \mathcal{F}_{s_0}$$

, (x_1, y_1) $W^s(0)$, $W^u(0)$, s, u .

■

4. G_k^ρ :

$$G_{k+1}^\rho = \frac{1}{2}\sum_{m=1}^{k} \mathfrak{D}_m \mathfrak{D}_{k+1-m} g^\rho + \sum_{m=1}^{k} \mathfrak{D}_m \hat{G}_{k+1-m}^\rho,$$

$\hat{G}_{k+1}^\rho$ G_{k+1}^ρ :

$$G_k^\rho = \sum_{j=2}^{k} \frac{\mathcal{L}_k^{(j)} g^\rho}{j!},$$

$$\hat{G}^{\rho}_{k} = \sum_{j=2}^{k} \frac{\mathcal{L}^{(j)}_{k} g^{\rho}}{(j+1)!},$$

$$\rho \in \lambda, \Lambda, x, y.$$

:

$$\mathcal{L}^{(j)}_{k+1} g^{\rho} = \sum_{m=1}^{k+2-j} \mathfrak{D}_m \mathcal{L}^{(j-1)}_{k+1-m} g^{\rho}$$

$$G^{\rho}_{k+1} = \sum_{j=2}^{k+1} \frac{\mathcal{L}^{(j)}_{k+1} g^{\rho}}{j!} = \sum_{j=2}^{k+1} \sum_{m=1}^{k+2-j} \frac{\mathcal{L}^{(j-1)}_{k+1-m} g^{\rho}}{j!} = \sum_{m=1}^{k} \mathfrak{D}_m \sum_{j=2}^{k+2-m} \frac{\mathcal{L}^{(j-1)}_{k+1-m} g^{\rho}}{j!} =$$

$$= \sum_{m=1}^{k} \mathfrak{D}_m \sum_{l=1}^{k+1-m} \frac{\mathcal{L}^{(l)}_{k+1-m} g^{\rho}}{(l+1)!} = \sum_{m=1}^{k} \mathfrak{D}_m \left(\sum_{l=2}^{k+1-m} \frac{\mathcal{L}^{(l)}_{k+1-m} g^{\rho}}{(l+1)!} + \frac{1}{2} \mathcal{L}^{(1)}_{k+1-m} g^{\rho} \right) =$$

$$= \sum_{m=1}^{k} \mathfrak{D}_m \left(\hat{G}^{\rho}_{k+1-m} + \frac{1}{2} \mathfrak{D}_{k+1-m} g^{\rho} \right) \blacksquare$$

5. 1) $\lambda_j, \Lambda_j, x_j, y_j, \quad j \le k-1 \quad \tau$,
$G^x_k(\tau), \tilde{G}^x_k(\tau), G^{\Lambda}_k(\tau), G^y_k(\tau)$. $G^x_k(\tau), \tilde{G}^x_k(\tau), G^{\Lambda}_k(\tau)$, $G^y_k(\tau), \tilde{G}^y_k(\tau)$ $G^x_k(\tau)$.

2) $G^x_2(\tau), \tilde{G}^x_2(\tau), G^{\Lambda}_2$ - , $G^y_2(\tau), \tilde{G}^y_2(\tau)$ - .

:

1) . .. $x_0(\tau)$ – , $y_0(\tau), \lambda_0(\tau)$ – :

$$g^{\Lambda}|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} = U(x_0, y_0) \sin\lambda_0 - V(x_0, y_0) \cos\lambda_0 \quad \text{- ,}$$

$$g^{x}|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} = -\varepsilon \left(2F y_0 - \frac{\partial U}{\partial y} \cos\lambda_0 - \frac{\partial V}{\partial y} \sin\lambda_0 \right) \quad \text{- ,}$$

$$g^{y}|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} = \varepsilon \left(2F(x_0 + \hat{x}_0) + e_J G - \frac{\partial U}{\partial x} \cos\lambda_0 - \frac{\partial V}{\partial x} \sin\lambda_0 \right) \quad \text{- .}$$

:

- (x_0) .
- $(y_0 \quad \lambda_0)$.

, $+1$ -1 - , :

$$\begin{aligned}
&\left. \frac{\partial^{p+q+m} g^{\Lambda}}{\partial x^p \partial y^q \partial \lambda^m} \right|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} \quad \Leftrightarrow \quad -(-1)^{q+m}, \\
&\left. \frac{\partial^{p+q+m} g^{x}}{\partial x^p \partial y^q \partial \lambda^m} \right|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} \quad \Leftrightarrow \quad -(-1)^{q+m}, \\
&\left. \frac{\partial^{p+q+m} g^{y}}{\partial x^p \partial y^q \partial \lambda^m} \right|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} \quad \Leftrightarrow \quad (-1)^{q+m}.
\end{aligned} \tag{57}$$

, $G^{\Lambda}_k(\tau), G^x_k(\tau), G^y_k(\tau)$:

$$G^{\Lambda}_k(\tau) = \sum_{p_1, q_1, m_1, \ldots} \left. \frac{\partial^{p+q+m} g^{\Lambda}}{\partial x^p \partial y^q \partial \lambda^m} \right|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} (x_{p_1} y_{q_1} \lambda_{m_1} \ldots),$$

$$G_k^x(\tau) = \sum_{p_1,q_1,m_1,\ldots} \frac{\partial^{p+q+m} g^x}{\partial x^p \partial y^q \partial \lambda^m}\bigg|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} (x_{p_1} y_{q_1} \lambda_{m_1} \ldots),$$

$$G_k^y(\tau) = \sum_{p_1,q_1,m_1,\ldots} \frac{\partial^{p+q+m} g^y}{\partial x^p \partial y^q \partial \lambda^m}\bigg|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} (x_{p_1} y_{q_1} \lambda_{m_1} \ldots),$$

, .

, (.. $\leq k-1$), $G_k^\Lambda(\tau), G_k^x(\tau), G_k^y(\tau)$. , g^Λ g^x , g^y , .
$\tilde{G}_k^x, \tilde{G}_k^y$ $G_k^\Lambda(\tau), G_k^x(\tau), G_k^y(\tau)$.

2) , $x_0(\tau), y_1(\tau), \tilde{x}_1, \lambda_1$ – , $y_0(\tau), x_1(\tau), \tilde{y}_1$ – .

$$G_2^\rho(\tau) = \sum_{p+q+m=2,\quad p,q,m\geq 0,\quad p,q,m\neq 2} \frac{\partial^2 g^\rho}{\partial x^p \partial y^q \partial \lambda^m}\bigg|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} x_1^p y_1^q \lambda_1^m \tag{58}$$

, $x_1^p y_1^q \lambda_1^m$ $(-1)^p$. , (57), (58) $-(-1)^{p+q+m}$ $\rho \in \{\Lambda, x\}$ $(-1)^{p+q+m}$ $\rho = y$. $p+q+m=2$, :
$G_2^x(\tau), G_2^\Lambda$ - , $G_2^y(\tau)$ - .
$\tilde{G}_2^x, \tilde{G}_2^y$ $G_2^\Lambda(\tau), G_2^x(\tau), G_2^y(\tau)$. ■